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3D reconstruction and segmentation of grape bunches for robotic berry thinning

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ABSTRACT

Table grape thinning requires selective removal of 20–40% of berries from dense clusters. In practice, workers decide which berries to remove by considering both the approximate berry count and local 3D spatial characteristics such as crowding and relative positioning. Automating this task is challenging because conventional 2D image-based approaches suffer from occlusion-related counting errors and lack explicit 3D spatial information necessary for reliable manipulation. We propose a robot-integrated vision pipeline that reconstructs grape bunch structure from posed multi-view RGB images. Neural Radiance Fields (NeRF) is used to learn a volumetric scene representation, from which a dense, low-noise point cloud is extracted via depth back-projection, and RANSAC-based geometric fitting models individual berries and stems, enabling berry-level segmentation and orientation estimation for manipulation planning. The perception pipeline uses an eye-in-hand RealSense D405 camera mounted on a Fanuc CRX-5iA collaborative robot. Camera poses are derived from the robot kinematic chain, allowing the reconstructed point cloud and detected berry centers to be expressed directly in the metric robot base frame without external scale recovery. In robot-mounted RealSense D405 experiments on 10 grape bunches, RANSAC sphere fitting achieved a counting MAE of 0.50 berries, RMSE of 0.71 berries, and mean center localization error of 2.71 mm. On a 52-bunch benchmark, RANSAC sphere fitting outperforms the learning-based SoftGroup++ method for 3D berry instance segmentation (92.1% vs 82.8% average precision) without requiring annotated training data. In 35 manipulation trials, the system achieved an 85.7% pre-grasp reachability rate and an 83.3% conditional target success rate demonstrating an end-to-end pipeline from neural reconstruction to manipulation-ready berry poses.

1. Introduction

Berry thinning is a critical horticultural practice in table grape cultivation that involves strategic removal of berries during early bunch development to improve fruit quality for better market value. This operation provides adequate spacing for remaining berries to expand, improves airflow within the bunch to reduce disease pressure from pathogens, and optimizes bunch compactness for premium market presentation. The impact of proper berry thinning on final product quality is substantial and multifaceted. Berry thinning prevents excessive bunch compactness which is one of the most important physical barriers to avoid fungal infections (Roberto et al., 2015). Thinning treatments have been shown to increase berry weight, total soluble solids, sugar content (fructose and glucose), berry firmness, and mineral contents, while decreasing titratable acidity and improving the ratio of sugar to acid (Jia et al., 2023). The timing of thinning is equally critical. Early berry thinning (7 days after flowering) is most effective in reducing bunch compactness and achieving optimal bunch quality (Ivanišević et al., 2020; Roberto et al., 2015). Research has demonstrated that appropriate thinning reduces grape yield but increases mean cluster weight, total phenols, anthocyanins, and individual berry size, while the bunch becomes more open and less inclined toward disease development (Karoglan et al., 2014). Beyond fruit quality, controlled bagging treatments combined with proper thinning timing can effectively regulate light exposure, influencing sugar metabolism and anthocyanin accumulation during berry maturation (Zha et al., 2019). These quality improvements directly affect market price, making berry thinning economically essential despite its labor intensity.

However, berry thinning faces critical operational constraints in commercial production. The suitable period for this operation is limited

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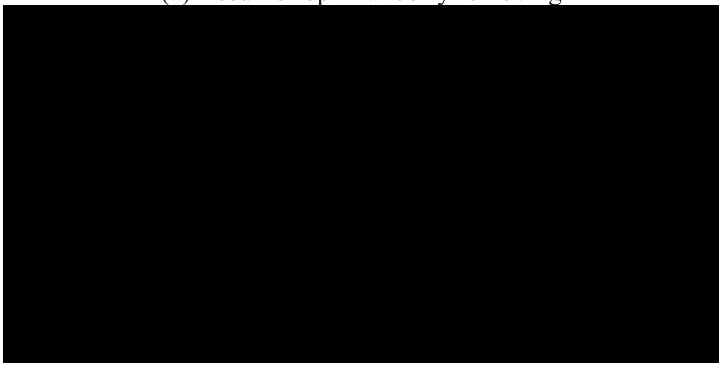
to only one to two weeks post-flowering, when there is still enough space among berries to allow unnecessary berries to be cut without damaging those to be retained, before the grapes begin to accumulate sugar. This narrow time window, combined with labor shortages and aging farming populations in major production regions such as Japan, creates severe bottlenecks in table grape production. In Japan, technologies enabling unskilled farmers to perform berry thinning using deep learning and augmented reality technologies are reported (Buayai et al., 2023b).

The practice of high-quality berry thinning by skilled farmers relies on understanding both the spatial arrangement and developmental potential of berries within the bunch. Accurate berry counting serves as a key information for effective thinning decisions. Farmers must first access the initial berry count (typically 60–80 berries in early bunch development) to determine the removal target (typically 35–40 berries for premium table grapes) based on variety-specific quality standards. As illustrated in Fig. 1a, the temporal transformation from pre-thinning to harvest demonstrates why geometric and developmental assessment is critical: an initial bunch of 70 berries at early development (left) undergoes strategic thinning during the narrow 2–3 weeks post-flowering window to approximately 35 berries (middle), which then expand over 2 months to produce marketable full-grown bunch (right). This progression reveals that farmers must assess not only the current berry count but also predict the 3D spatial arrangement evaluating whether remaining berries will have adequate spacing to expand without physical interference from neighbors. The target bunch architecture after thinning (Fig. 1b) follows a hierarchical structure: a total length of 7–8 cm with berries distributed across multiple lateral sub-branches, typically maintaining 4–5 berries per 2 sub-branches in the upper region, 3 berries per 5 sub-branches in the middle section, and 2 berries per 5 sub-branches in the lower portion, yielding target counts of 28–40 berries. This spatial organization serves multiple agronomic objectives: adequate spacing allows each berry to expand to target marketable size without physical interference from neighbors, improved airflow between berries reduces humidity and fungal disease pressure, and the open architecture facilitates uniform light exposure promoting anthocyanin development and sugar accumulation that characterizes premium table grapes. This geometric reasoning about 3D structure, spacing adequacy, and growth potential cannot be reliably inferred from single 2D images due to occlusion and perspective ambiguity. For robotic execution, this task further demands accurate 3D localization of individual berries in metric coordinates compatible with robot motion planning, along with collision-aware approach.

2D image-based methods achieve fast berry detection and counting using deep learning with hand-crafted features or end-to-end neural networks (Buayai et al., 2021; Buayai et al., 2023b; Du & Liu, 2023; Woo et al., 2023a). Early work employed regression model based on hand crafted features (Buayai et al., 2021), while recent approaches use convolutional neural networks for predicting berry counting from single image (Woo et al., 2023a) or instance segmentation using advanced architectures like Swin transformer with adaptive spatial feature fusion (Du & Liu, 2023). While instance segmentation methods like AS-SwinT achieve impressive 2D detection performance (AP $>$ 90%), these methods suffer from fundamental limitations that make them inadequate for precision berry thinning. First, they exhibit view-dependent predictions: the same bunch yields different berry counts from different viewpoints due to occlusion. Reported counting errors increase systematically with bunch density, reaching mean absolute errors of 3.72 and 5.46 berries for bunch containing 52–61 and more than 61



(a) Result of optimal berry removing



(b) Remaining criteria of grape berries

Fig. 1. Berry thinning process and target bunch structure for Shine Muscat table grape. (a) Temporal progression from pre-thinning to post-thinning, and the resulting full-grown bunch. (b) Target post-thinning structure.

berries respectively (Woo et al., 2023a), with RMSE of 4.72–7.13 berries for larger, denser bunches (Du & Liu, 2023). This error magnitude is problematic for berry thinning, where the task typically involves removing berries from initial counts of 70 down to target counts of 35–40. More critically, pixel coordinates from 2D detection cannot directly guide robot motion without depth information. Extensions to berry removal recommendation (Buayai et al., 2023b; Tamura et al., 2024) operate in image space by training a CNN model with bunch images before and after thinning performed by a skilled farmer. However, due to lacking 3D depth and structural information, such methods cannot provide geometric reasoning about actual 3D inter-berry distances and collision-free approach paths.

Accurate 3D positioning for robotic manipulation requires depth information that 2D method cannot provide. Single-view depth sensing approaches attempted to bridge this gap by integrating RGB-D cameras or stereo vision for 3D positioning, Buayai et al. (2023a) combined depth sensing technology with neural networks to control robotic arm end-effector for grape berry approaching, integrating 2D berry detection with depth maps to estimate 3D target coordinates and training neural networks to map detected berry positions to feasible robot end-effector poses. However, single-view depth sensing achieves only 20% approach success due to challenges for outdoor agricultural applications: noisy depth estimates under variable lighting. Furthermore, single-view depth provides no geometric consistency across viewpoints, depth estimates for the same berry differ depending on viewing angle and cannot handle severe occlusion where berries are completely hidden from the sensor's line of sight. These depth errors propagate directly to collision checking and trajectory planning limiting reliability for safe manipulation in cluttered bunches.

Multi-view reconstruction methods offer the potential for complete geometric understanding. Classical Structure-from-Motion (SfM) and Multi-View Stereo (MVS) have been applied to grape bunch reconstruction (Woo et al., 2023b). However, photometric stereo matching faces significant challenges for grape berries. This approach relies on identifying corresponding texture features across multiple views. Grape berries present two key challenges for this approach: (1) surface texture-poor that provides few distinctive matching points, and (2) small physical size (typically 10–20 mm diameter) relative to practical reconstruction resolution. These factors result is sparse, incomplete reconstruction where many berries are missed or only partially captured.

Neural Radiance Fields (NeRF) (Mildenhall et al., 2020) overcome these texture-matching limitations through learned volumetric representations from multi-view images to render photorealistic images from arbitrary viewpoints. Unlike classical MVS methods that relies on explicit feature matching and produce sparse point clouds, NeRF learns continuous volumetric scene representations without requiring explicit depth supervision (Li et al., 2025). Recent agricultural applications demonstrate NeRF's capability for high-fidelity crop reconstruction of small and complex crop structures: rice panicles achieving mean F1 score of 86.9% and IoU of 79.8% (Yang et al., 2024), cherries (Hu et al., 2024), general plant phenotyping (Smitt et al., 2023), and tomato morphological analysis (Choi et al., 2024).

However, existing agricultural NeRF works focus on visualization and phenotyping rather than robotic manipulation, revealing critical gaps when extended to manipulation tasks. Three fundamental limitations persist: 2D mask fusion approaches (Meyer et al., 2024; Smitt et al., 2023) project neural network detections onto reconstructed geometry, inheriting view-dependent errors and requiring heuristic merging rules that introduce inconsistencies when berries appear differently across views. Shrestha et al. (2025) render novel views for training pose estimation networks but require separate networks per object category and provide no geometric primitives for collision checking. Phenotyping pipelines (Choi et al., 2024) require separate calibration procedures with known-dimension reference objects to recover metric scale and, critically, do not validate end-to-end manipulation.

NeRF was originally proposed for novel view synthesis by learning a continuous volumetric scene representation from multi-view images to render photorealistic images from arbitrary viewpoints. However, the learned density field implicitly encodes scene geometry, enabling extraction of dense 3D point clouds through ray-based depth estimation and back-projection. We leverage this geometric byproduct for downstream segmentation and manipulation rather than rendering capability itself. Critically for our application, NeRF's volumetric learning does not require explicit feature matching between views, making it well suited to texture-poor grape surfaces where classical multi-view stereo methods produce sparse, incomplete reconstructions. By extracting individual berry instances from reconstructed point cloud requires segmentation. For instance-level segmentation of 3D point clouds, two primary approaches are commonly adopted: learning based approaches and geometry-based approaches. Learning-based 3D instance segmentation method such as SoftGroup (Vu et al., 2022) directly segmenting 3D point clouds using neural networks. These methods require extensive manually annotated 3D training data, which is costly to produce. Furthermore, when berries are small and densely clustered with significant occlusion, learned segmentation can fail to reliably separate individual instances.

In contrast, geometry-based approaches rely on explicit shape priors and parametric primitive fitting. In this work, we adopt a RANSAC based geometric primitive fitting strategy that leverages the strong prior that grape berries are approximately by spherical. The method exploits the geometric constraint: berry surfaces are well approximated by spheres, whereas stem structures are more accurately modeled by connected cylindrical primitives. This distinction enables robust separation through primitive fitting without dependence on annotated training data. Recent work has demonstrated RANSAC's robustness to imaging noise and outliers in agricultural point clouds, achieving $R^2 > 0.90$ for diameter and angle measurements even with up to 10% outlier contamination in structure-from-motion reconstructed data (Ghahremani et al., 2021). Prior applications have successfully identified individual grape berries as spherical primitives using laser scanning combined with RANSAC shape detection (Schnabel et al., 2007) in controlled settings (Mack et al., 2018).

This work builds on established components, including NeRF-based reconstruction, RANSAC primitive fitting, skeletonization, and robotic motion planning, and focuses on the integration into a robot-ready perception-to-manipulation pipeline for grape berry thinning. Existing agricultural NeRF studies mainly focus on visualization, trait measurement, or synthetic data generation, whereas robotic berry thinning requires relative berry positions to robot coordinate system, local stem-aware approach orientations, and collision-aware motion execution. To address these requirements, our system integrates robot-derived camera poses, NeRF-based volumetric reconstruction, depth-based point cloud extraction, RANSAC geometric primitive fitting, stem skeleton extraction, and motion planning into a complete pipeline that produces berry-level 3D positions and 6-DoF approach poses for robotic execution. The key insight is that neural volumetric reconstruction and geometric shape detection are complementary: NeRF learns a continuous scene representation from multi-view RGB images, from which a dense point cloud is extracted through depth rendering and back-projection, while RANSAC sphere fitting provides robust, training-free berry segmentation by exploiting spherical geometry priors. By deriving camera poses from robot kinematics, the extracted grape bunch point cloud is expressed directly in the metric robot base frame, enabling end-to-end validation from perception to physical manipulation.

Our specific contributions are:

1. **Robot-ready metric NeRF reconstruction for manipulation**: We develop an integrated eye-in-hand multi-view reconstruction framework in which RGB images from a robot-mounted RealSense D405, and camera poses derived from robot forward kinematics chain through ROS 2 TF2 are used for NeRF training. This produced

grape bunch point clouds directly in metric robot base coordinates system, avoiding external scale estimation or frame transformation and enabling seamless integration with downstream robot motion execution.

2. **Training-free 3D berry segmentation and structure-aware pose estimation:** We propose a geometry-driven pipeline that segments individual berries directly from NeRF-exported point clouds without annotated training data. RANSAC sphere fitting detects berry instances and estimates their centers and radii, while the residual point cloud is used to reconstruct the supporting stem and branch structure. This produces both berry-level instances and structure-aware 6-DoF target poses required for robotic thinning.
3. **System validation**: We validate the proposed pipeline on robot-derived metric reconstruction from 10 grape bunches and demonstrate accurate berry detection with an MAE of 0.50 berries, RMSE of 0.71 berries, and mean berry-center localization error of 2.71 mm. We further evaluate the geometry-based segmentation method on a 52-bunch high-resolution benchmark, where it outperformed SoftGroup++ without annotated training data. Physical robot manipulation experiments on 35 sequential berry targets showed that stem-aligned 6-DoF pose estimation improve collision-free grip success from 78.3% to 96.0% among target-reachable trials.

2. Materials and methods

This section presents our complete pipeline for grape bunch reconstruction and robotic manipulation through multi-view neural reconstruction and RANSAC fitting. Fig. 2 illustrates the five-stages approach: (1) multi-view data acquisition using an eye-in-hand camera mounted on industrial manipulator (Section 2.2), (2) NeRF-based 3D reconstruction trained from 360-degree view images with metric-scale preservation through robot-derived camera poses (Section 2.1), (3) RANSAC-based primitive detection to segment individual berries and stems in the point cloud (Section 2.3), (4) grape bunch structure extraction from extracted primitives and 6-DoF berry pose optimization that aligns end-effector approach direction with berry-stem attachment geometry (Section 2.4), (5) robot motion planning and execution using optimized berry poses as manipulation targets (Section 2.5).

2.1. 3D reconstruction using NeRF

NeRF represents a 3D scene as a continuous function $f_\theta : (x, d) \rightarrow (c, \sigma)$ mapping 3D position $x \in R^3$ and viewing direction $d \in S^2$ to RGB color $\boldsymbol{c}$ and volume density $\sigma \geq 0$. The function f_θ is parameterized by a multi-layer perceptron (MLP) with learnable weights θ. To render the color observed along camera ray, we cast a ray $r(t) = \boldsymbol{o} + t\boldsymbol{d}$ from camera center $\boldsymbol{o}$ through a pixel in direction $\boldsymbol{d}$, sample N points along the ray depths at $\{t_i\}_{i=1}^{N}$ between near plane t_{near} and far plane t_{far}, and compute:

$$\widehat{\boldsymbol{C}}(\boldsymbol{r}) = \sum_{i=1}^{N} T_i \alpha_i \boldsymbol{c}_i, \text{where}(\mathbf{c}_i, \sigma_i) = f_\theta(\boldsymbol{r}(t_i), \boldsymbol{d} \tag{1}$$

The transmittance T_i represents the probability that ray reaches sample point i without being occluded, while the alpha compositing weight α_i determines each sample's contribution to the final color:

$$\alpha_i = 1 - exp(-\sigma_i \Delta_i), \Delta_i = t_i + 1 - t_i \tag{2}$$

$$T_i = \exp\left(-\sum_{j=1}^{i-1} \sigma_j \Delta_j\right) = \prod_{j=1}^{i-1} (1 - \alpha_j) \tag{3}$$

Training optimizes the weights θ to minimize photometric reconstruction loss between rendered colors and observed pixel colors across all training views:

$$\mathscr{L}_{photo} = \sum_{\mathrm{r} \in \mathrm{R}} \|\widehat{C}(\mathrm{r}) - C(r)\|^2 \tag{4}$$

where R is the set of all rays sampled from the training images and $C(r)$ is the ground truth pixel color.

In this study, Nerfacto method from the Nerfstudio framework (Tancik et al., 2023) was trained with 5000 iterations, this number of iterations was evaluated the trade-off between quality and training time (result in Section 3.1), a learning rate of 0.01 for the networks (0.001 for camera optimization), and a batch size of 4096 rays. Near and far planes set to $t_{near} = 0.05$ m and $t_{far} = 15$ m.

After NeRF training, a dense point cloud is extracted through ray-based depth sampling and back-projection. Rather than rendering full depth images for each camera view, we repeatedly sample batches of training rays from the NeRF data manager. For each ray $\mathrm{r}(\mathrm{t}) = \mathbf{o} + t\mathbf{d}$, the model predicts color, opacity, and depth via volumetric rendering. The expected depth is computed as the transmittance-weighted average

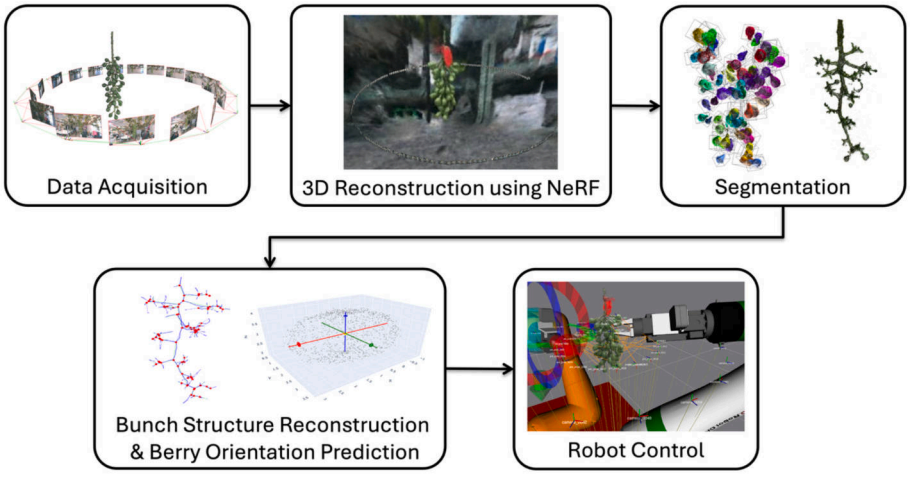


Fig. 2. Pipeline overview. Multi-view images captured from an eye-in-hand camera configuration provide both RGB appearance and metric camera poses derived from robot kinematics (Section 2.2). These poses train a NeRF model that renders novel views and exports dense point clouds directly in the robot base coordinate frame (Section 2.1). RANSAC sphere detection identifies individual berries and extracts their 3D centers and radii (Section 2.3). The residual point cloud after berry removal undergoes Laplacian-based contraction to extract a skeletal representation of the stem structure and each detected berry's 6-DoF pose is optimized to align the end-effector approach direction with the local stem attachment direction inferred from the skeleton (Section 2.4). The optimized poses serve as motion planning targets for collision-checked robot approaching, with simplified collision geometry representing the bunch volume, support poles, and workspace boundaries (Section 2.5).

of sample depths:

$$\widehat{D}(\boldsymbol{r}) = \frac{\sum_{i=1}^{N} T_i \alpha_i t_i}{\sum_{i=1}^{N} T_i \alpha_i} \tag{5}$$

and a 3D point is obtained directly in the NeRF coordinate frame by back-projecting along the ray:

$$p = o + \widehat{D}(\boldsymbol{r})\boldsymbol{d} \tag{6}$$

Points are retained only if their rendered opacity exceeds a fixed threshold $\tau_{opacity} = 0.5$, which removes samples corresponding to empty space:

$$\text{opacity}(\mathbf{r}) = \sum_{i=1}^{N} T_i \alpha_i \tag{7}$$

To further improve geometric quality, statistical outlier removal is applied using 20 nearest neighbors and a standard deviation ratio of 10.0. The final-colored point cloud is exported in PLY format and contains approximately two million points per grape bunch.

NeRF training thus requires two inputs: a set of multi-view RGB images with sufficient viewpoint coverage and per-image camera pose estimates comprising intrinsic parameters and 6-DoF extrinsic transformations. The following section describes how these inputs are obtained through our robotic data acquisition system.

2.2. Data acquisition

The images used to train the NeRF model for the manipulation pipeline were captured with an Intel RealSense D405 camera mounted on a Fanuc CRX-5iA robot with an eye-in-hand configuration. This arrangement supports repeatable viewpoint sampling and provides a consistent relationship between camera and the robot end-effector, which is required to express reconstructed geometry in the robot base frame.

2.2.1. Image capture

RGB images were recorded at a resolution of 840 × 480 pixels using only the RGB sensor of an Intel RealSense D405 mounted on Fanuc CRX-5iA with custom end-effector (Fig. 3a). The D405 depth channel was not used for NeRF reconstruction or point-cloud generation. To ensure photometric consistency across viewpoints and minimize auto-exposure artifacts that degrade NeRF training quality, camera settings were held fixed during acquisition: ISO was set to 100, auto-exposure was disabled with a fixed exposure time of 20 ms, and white balance was locked to 4000 K to compensate for greenhouse lighting conditions.

The robot was commanded through a complete 360° circular trajectory around each target grape bunch. The trajectory was planned in the robot joint space, with the camera positioned at a working distance of approximately 300 mm from the bunch center. Total of 279 images were captured per acquisition sequence, corresponding to 1.29° angular increment between consecutive frames. This sampling density provides substantial overlap between consecutive views and was further evaluated through the view-sampling ablation study reported in Section 3.2.

During acquisition, each saved RGB image was paired with camera pose obtained from the robot kinematic chain through ROS 2 TF2 at the time of saving (Fig. 3b). The transform base link to D405 frame was converted to the OpenGL camera convention allowing NeRF reconstruction to be expressed directly in the robot base coordinate frame. The resulting posed image sequence therefore allowed NeRF reconstruction to be expressed directly in the robot base coordinate frame. Fig. 3c shows a representative circular D405 acquisition trajectory and the corresponding camera viewpoints around the target bunch.

Although the proposed reconstruction pipeline does not use active depth sensing, RGB image quality remains important because NeRF optimization depends on multi-view photometric consistency. The effect of RGB acquisition quality was therefore evaluated separately using a paired robot-mounted D405/iPhone comparison, as reported in Section 3.4. The D405 depth channel may still be useful in future deployment for coarse target-bunch localization, initial bunch-center estimation, scan-trajectory initialization, ROI placement, and workspace-level collision awareness.

2.2.2. Camera intrinsics and extrinsic

NeRF reconstruction requires camera intrinsics parameters and per-image extrinsic poses. Camera intrinsics were obtained from the RealSense D405 factory calibration provided by the ROS 2 camera driver node. The pinhole intrinsic matrix K (focal lengths and principal points):

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where $\left(f_x, f_y\right) = (433.3, 432.2)$ pixels are the focal lengths, and $(c_x c_y) = (848.0, 480.0)$ pixels are the principal points. Images were processed at the original resolution (840 × 480), so intrinsics were applied without additional scaling.



(a) Robot-mounted Intel RealSense D405 setup (b) Robot TF2 transform chain (c) D405 acquisition trajectory and camera poses

Fig. 3. Eye-in-hand Intel RealSense D405 acquisition setup and robot-derived camera poses. (a) Intel RealSense D405 mounted on the Fanuc CRX-5iA end-effector for robot-integrated RGB image acquisition. (b) ROS 2 TF2 transform tree used to obtain the camera pose from the robot kinematic chain. (c) Representative 360° circular image acquisition trajectory around the target grape bunch. The small image planes indicate camera viewpoints along the trajectory. The D405 RGB images and TF2-derived camera poses were used for NeRF reconstruction in the robot base coordinate frame.

Camera extrinsic poses, the 6-DoF position and orientation of the camera for each captured image are conventionally estimated using structure-from-motion (SfM), with COLMAP (Schonberger & Frahm, 2016) being the de facto preprocessing tool in NeRF workflows. COLMAP performs feature extraction, cross-view matching, and bundle adjustment to recover camera poses from image correspondences. However, this offline processing adds computational overhead (typically several minutes for hundreds of images) and, more critically, produces pose estimates in an arbitrary coordinate frame with unknown metric scale. In our robotic pipeline, we bypass COLMAP entirely by deriving camera poses directly from robot's kinematic chain via the ROS 2 TF2 (Transform Framework). For each captured image, its timestamp was recorded, and the corresponding camera pose was queried from TF2 at that timestamp by computing the transform chain from the robot base frame to the camera optical frame. This approach provides two advantages: (1) it eliminates the computational cost and potential failure modes of feature-based pose estimation, and (2) it produces camera poses in the metric robot base coordinate system, enabling the reconstructed point cloud to be used directly for robot motion planning without additional frame transformations or scale calibration.

2.2.3. Environment filtering

Before berry detection, we remove background points outside the region of interest to eliminate false positive detection from environmental clutter. The NeRF reconstruction produces a dense point cloud of entire visible scene, including the target grape bunch, neighboring bunch, canopy, and distance object.

Our filtering strategy leverages the known camera trajectory geometry from eye-in-hand acquisition setup. During data acquisition the camera was positioned at a fixed working distance of 300 mm from the estimated bunch center throughout the 360° circular trajectory. We define a spherical region of interest (ROI) centered at this target position with radius of 200 mm (diameter 400 mm). This radius was chosen to provide sufficient margin beyond typical grape bunch dimension.

As shown in Fig. 4, this spherical filtering effectively isolates the grape bunch while removing the environment objects in the raw reconstruction and provides a cleaner input for RANSAC-based berry detection. This design assumes that the target bunch has been selected before reconstruction and that the objective is to recover berry-level geometry for one bunch at a time. Multi-bunch and canopy-level reconstruction would require additional target-bunch detection, bunch-level instance separation, and adaptive scan planning, which are discussed as future work in Section 4.

2.3. Berry segmentation using RANSAC geometry fitting

The geometric characteristics of grape bunch point clouds motivate the adoption of RANSAC-based sphere detection for berry segmentation. As reported by Mack et al. (2018), berry surfaces exhibit approximately spherical and smoothly curved geometry in laser-scanned point clouds, whereas stem structures are more accurately described by connected cylindrical primitives. This distinction in local surface geometry makes geometric primitive fitting well suited for separating berries from stem structures. In particular, the spherical shape of berries supports robust detection through sphere fitting, while the geometry of stems is naturally excluded under appropriate geometric constraints.

Grape berries are approximately spherical and therefore amenable to detection via geometric primitive fitting. We employed the RANSAC algorithm for robustly identify multiple spherical structures in the presence of noise and outliers (Schnabel et al., 2007). Each sphere is parameterized by a center $\mathrm{c} \in \mathrm{R}^3$ and a radius $r > 0$. The algorithm operates iteratively by randomly sampling minimal point sets from the point cloud, fitting sphere hypotheses to these samples, and determining the subset of points that support each hypothesis. Sphere hypotheses are generated by randomly sampling four points from the point cloud and computing an initial estimate of the sphere center $\mathrm{c} = (\mathrm{x}, \mathrm{y}, \mathrm{z})^\top$ and radius r. For each hypothesis, an inlier set $\mathscr{I}$ is defined as the set of points whose distance to the sphere surface is below a predefined tolerance ϵ. A point p is classified as an inlier if $|\|\mathrm{p} - \mathrm{c}\| - r| \leq \epsilon$. If the number of inliers exceeds a minimum threshold $\mathrm{N}_{\min}$ and $r_{min} \leq r \leq r_{max}$, the sphere parameters are refined using least-squares fitting over all inliers and the hypothesis is accepted; otherwise, it is rejected. This procedure is repeated for a maximum of $K_{\max}$ iterations to identify the sphere with the largest inlier support. Since grape bunch point clouds typically contain multiple berries, sphere detection is performed iteratively. After each successful detection, the corresponding inlier points are removed from the point cloud, and the detection process is applied to the remaining points. Iteration continues until either an insufficient number of points remains, or no further valid sphere hypotheses can be identified. To avoid biologically implausible detections, candidate spheres are constrained to an admissible radius range $[r_{min}, r_{max}]$ derived from known berry sizes of the target grape observed from the dataset in Section 3.5.1. These parameters are summarized in Table 1.

2.4. Bunch structure reconstruction and berry orientation prediction

Robotic berry thinning requires an accurate 6-DoF target pose for each berry so that the end effector can align its tool axis with the intended approach direction and maintain clearance from neighboring

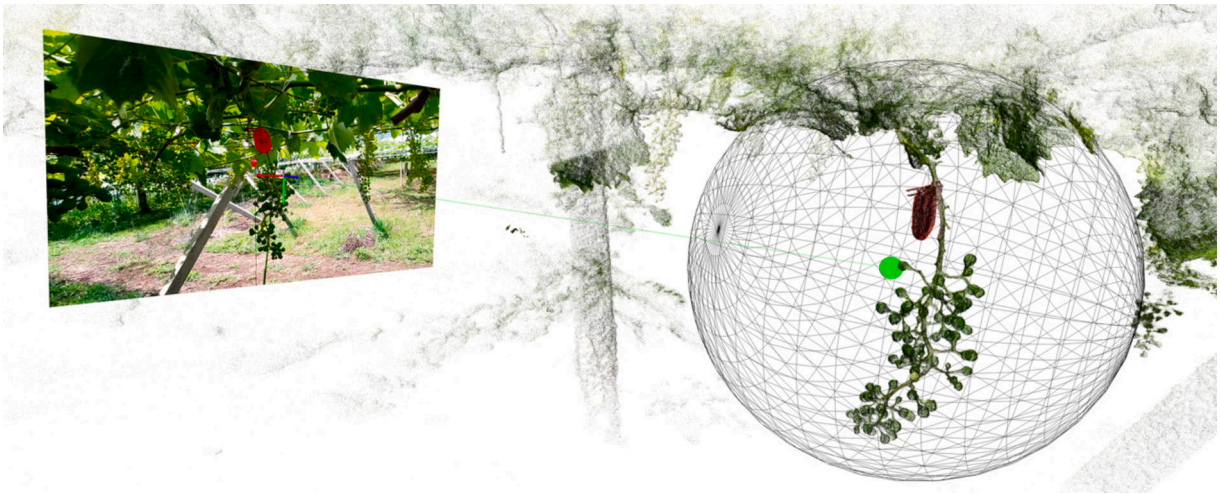


Fig. 4. Spherical region of interest filtering for background removal.

Table 1
RANSAC sphere detection parameters.

Parameter	Symbol	Value	Description
Distance tolerance	ϵ	1.5 mm	Maximum deviation of points from fitted sphere surface
Radius range	$[r_{min}, r_{max}]$	[10,20] mm	Berry radius bounds from observed grape sizes
Minimum inliers	N_{min}	400 points	Minimum supporting points required per sphere hypothesis
Maximum iterations	K_{max}	1000	Maximum RANSAC iterations per detection cycle
Normal angle threshold	θ_{max}	25°	Maximum angular deviation between sphere and point normals

berries and the supporting branch structure. In dense grape bunches, a 3D berry center coordinate alone is insufficient for reliable manipulation because it defines only the position of the target, leaving end-effector orientation unconstrained. As illustrated in Fig. 5a, a target berry located deep inside a dense cluster cannot be reliably approached using only its 3D center coordinate. Fig. 5b shows that without berry orientation to its local sub-branch direction may cause the gripper to collide with neighboring berries during insertion. Only orientations whose approach axis aligns with the local sub-branch direction, shown in Fig. 5c, allow safe tool insertion without disturbing the surrounding cluster. We therefore estimate a manipulation-ready pose for each detected berry by leveraging the spatial relationship between the berry surface and its stem attachment. The procedure consists of two steps. First, we extract a branch and stem skeleton from the residual point cloud obtained after removing berry inliers from the RANSAC sphere fitting stage. Second, we optimize each berry's orientation such that its approach axis is consistent with the local stem attachment direction inferred from skeleton. The resulting rotation matrix, together with the berry center, defines 6-DoF target pose used to compute a pre-grasp pose and subsequent approach motion.

2.4.1. Grape bunch skeletonization

To recover the supporting structure of the bunch, we apply skeletonization to the residual point cloud $C \in \mathbb{R}^{N_C \times 3}$ (Fig. 6a), where N_C is the number of remaining points after excluding all berry inliers. We employed Laplacian-based contraction (LBC) (Cao et al., 2010) to iteratively contract C into a set of skeleton points in Fig. 6b. The LBC output can be discontinuous due to occlusion, varying point density, and partial incompleteness, we further impose connectivity by constructing a minimum spanning tree (MST) over the skeleton points (Meyer et al., 2023). In Fig. 6c, the resulting graph provides a coherent representation of the branch and stem structure and serves as a geometric reference for estimating berry attachment directions in the subsequent orientation optimization.

2.4.2. Berry orientation optimization

Given each detected berry instance $\boldsymbol{P_j}$ with center $\mathbf{c_j}$, our goal is to estimate a rotation R that aligns the end-effector approach axis with the berry-to-stem attachment direction. We first compute an initial rotation R_j^{PCA} from PCA on $\boldsymbol{P_j}$. The eigenvectors define three principal axes ordered by decreasing variance; for approximately spherical berries, the third eigenvector (minimum variance direction) provides a geometric approximation of the berry's polar axis and serves as our initialization for the approach direction. The resulting initial berry orientation frames are visualized in Fig. 7d, showing the directional variance inherent in PCA-derived orientations. However, this initialization is often physical imprecise in practice, as can be assess visually. Furthermore, PCA alone provides no information about direction of the berry to local stem. We therefore refine this initialization by constraining the approach axis using the extract skeleton, which provides explicit geometric evidence of stem attachment direction.

Next, we identify the skeleton point closest to the berry center as a stem point:

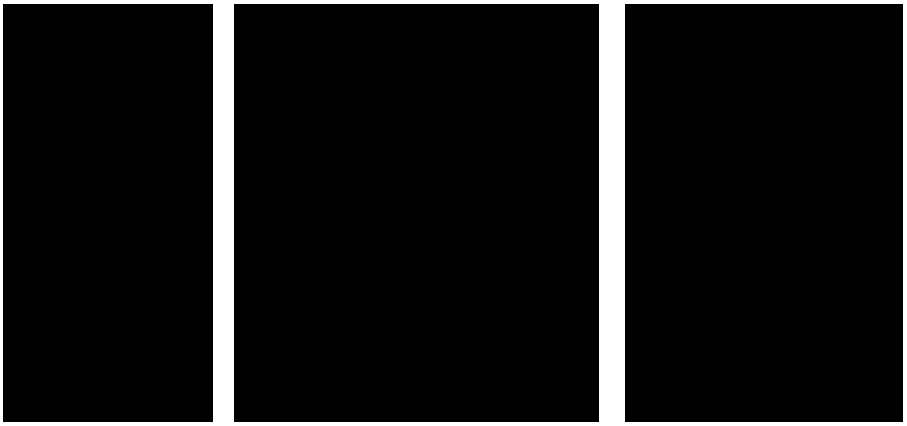
$$s_j^* = \underset{s \in S}{\operatorname{argmin}} \left\| \mathbf{s} - c_j \right\| \tag{8}$$

where S denotes the set of skeleton points obtained in Section 2.4.1. Fig. 7a illustrates this geometric relationship, showing the berry center $\mathbf{c_j}$, its nearest skeleton point s_j^* and the resulting stem direction vector d_j.

We then refine the berry orientation by aligning the third principal axis with the stem point s_j^*. The ideal approach direction is defined as the normalized vector from the berry center toward the skeleton stem point $d_j = \frac{s_j \bullet c_j}{\|s_j^* - c_j\|}$. As shown in Fig. 7b, the angular misalignment θ between the PCA-derived approach axis e_3 and the target stem direction d_j motivates this refinement step. The optimization objective is:

$$\mathscr{L}(R) = \arccos\left(< Re_3, d_j > \right) \tag{9}$$

where $e_3 = [0,0,1]^1$ is the third principal axis and $< \bullet, \bullet >$ denotes inner product. This directly minimizes angular misalignment between rotated approach axis Re_3 and the stem direction. Fig. 7c shows the resulting

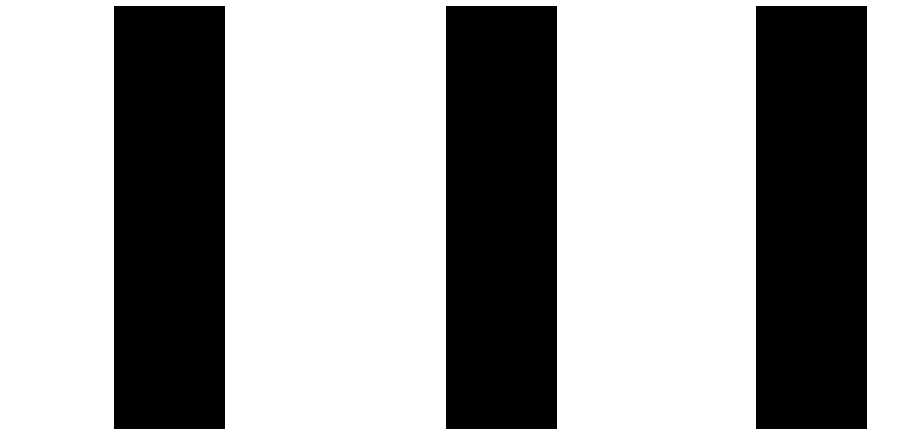


(a) Target berry deep inside a dense cluster.

(b) Fixed global orientation: approach axis misaligned, causing collision risk.

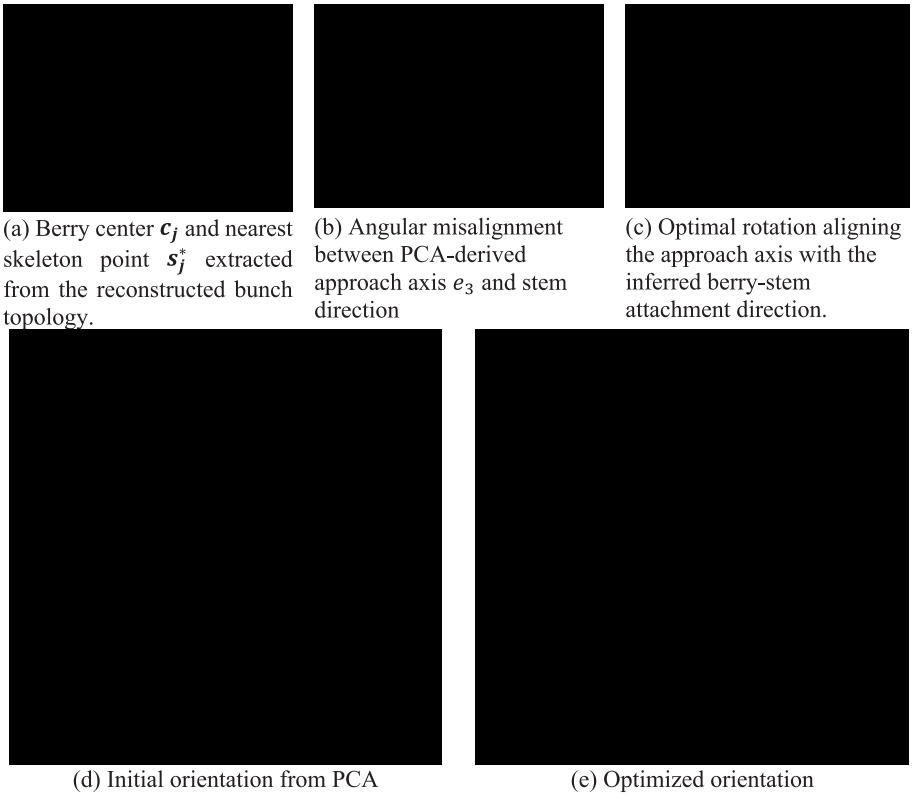
(c) Stem-aligned orientation: approach axis (blue) aligned with sub-branch direction.

Fig. 5. Motivation for 6-DoF pose estimation for berry manipulation. (a) Target berry located deep inside cluster, approaching correct angle is challenging for naïve 3D berry center. (b) The same berry targeted with geometrically reachable but orientation-unsafe end-effector poses; most approach axes cause the gripper to collide with neighboring berries during insertion. (c) The stem-aligned approach vector, clear path, and aligned the sub-branch direction.



(a) Original residual point cloud (b) Skeleton (c) Bunch topology

Fig. 6. Grape bunch skeletonization via Laplacian-based contraction (LBC). (a) Original residual point cloud after removing RANSAC berry inliers, representing the stem and branch structure. (b) Skeleton produced by iterative LBC contraction. (c) Final bunch topology obtained by connecting skeleton points via a minimum spanning tree (MST), providing a tree-structured geometric reference for berry-to-stem attachment estimation.



(a) Berry center $\boldsymbol{c_j}$ and nearest skeleton point $\boldsymbol{s_j^*}$ extracted from the reconstructed bunch topology.

(b) Angular misalignment between PCA-derived approach axis e_3 and stem direction

(c) Optimal rotation aligning the approach axis with the inferred berry-stem attachment direction.

(d) Initial orientation from PCA (e) Optimized orientation

Fig. 7. Structure-aware berry orientation inference using reconstructed grape bunch geometry. (a) Berry center c_j and nearest skeleton point s_j^* extracted from the reconstructed bunch topology, with the stem direction vector d_j illustrated. (b) Angular misalignment θ between the PCA-derived initial approach axis e_3 and the target stem direction, visualized per berry. (c) Optimal rotation aligning the approach axis with the inferred berry-stem attachment direction after gradient-based optimization. (d) Initial end-effector orientation frames at all detected berries derived from PCA, showing large directional variance. (e) Optimized end-effector orientation frames at all detected berries after 6D rotation optimization, showing consistent alignment with local stem directions.

optimal rotation, where the approach axis is fully aligned with the inferred berry-stem attachment direction.

To enable stable gradient-based optimization, we parameterized R using the continous 6D rotation representation of Zhou et al. (2019), which avoids the discontinuities and singularities inherent in Euler angles and axis-angle representations and yield smoother convergence and lower error as depicted in Fig. 24. We optimized the 6D parameters with Adam optimizer (Kingma & Ba, 2015) using a learning rate of 0.01 for up to 2000 iterations. Gradients are computed by automatic differentiation in PyTorch. The refinement typically reduces the alignment objective by approximately one order of magnitude relative to PCA initialization. Fig. 7e confirms this improvement at the bunch level: the optimized orientation frames are visibly more consistent with local stem directions compared to the PCA initialization in Fig. 7b. The final target poses for berry j is given (c_j, R_j), which is subsequently used to compute a pre-grasp pose and execute approach motions in the robot planning framework.

To evaluate the contribution of structure-aware orientation estimation, the stem-aligned approach is evaluated against a fixed-direction baseline in which the robot approaches each berry using an end-effector orientation with no geometric reasoning. Fig. 8 contrasts the spatial distribution of pre-grasp frames under two conditions. In Fig. 8a, without orientation constraints, all pre-grasp frames are clustered near the robot's home configuration, concentrating approach directions over a narrow angular range regardless of local berry geometry. In Fig. 8b, with stem-aligned orientation, the pre-grasp frames are distributed across a wider solid angle that matches the three-dimensional arrangement of the bunch's lateral branches. This structured diversity is a direct consequence of incorporating skeleton topology into the pose optimization and is the geometric prerequisite for the collision-free grip success rates reported in Section 3.7.2.

2.5. Robot control

Robot motion is executed by a ROS 2 motion node integrating TF2 for frame management, MoveIt 2 (Coleman et al., 2014) for motion planning and trajectory execution, and a gripper controller for end effector actuation. At initialization, the node loads the set of optimized berry poses computed in Section 2.4. Each berry instance j is represented by a target 6 DoF pose T_j^{target} and a corresponding pre grasp pose T_j^{pre}, both expressed in the robot base frame. These poses are published to the TF tree for verification in RViz.

Fig. 9 illustrates the manipulation pipeline. Fig. 9a shows the gripper coordinate frame definition. The gripper local z axis defines the approach direction and is aligned with the optimized berry orientation. During execution, the gripper approaches along the negative z direction, with the jaws-oriented perpendicular to the approach vector to minimize disturbance to neighboring berries.

Pre grasp poses are generated by offsetting 80 mm along the negative approach direction from each target pose. Fig. 9b visualizes the pre-grasp frames for an approach target berry. The diversity of orientations reflects the geometric optimization described in Section 2.4.2, where approach directions are aligned with local stem attachment structure rather than uniformly outward.

For each target berry j, the system executes a three-stage motion primitive:

Stage 1: Global motion to pre-grasp. Generate collision-aware trajectory from current configuration to pre-grasp pose T_j^{pre} (80 mm offset from target) using MoveIt 2 with OMPL RRTConnect planner (5 time-out). If planning fails, retry with alternative planners and relaxed tolerances.

Stage 2: Cartesian linear insertion. Execute straight-line motion from T_j^{pre} to T_j^{target}, enforcing direct approach near the berry.

Stage 3: Grasp. At target pose, actuate gripper close based on berry diameter, hold for 1 s for alignment verification, then open and retract.

2.6. Implementations

NeRF training and point cloud export were performed on a workstation with an AMD EPYC 7543P CPU (32 cores, 64 threads), with GPU acceleration provided by an NVIDIA A100 (80 GB PCIe). The system used NVIDIA driver 565.57.01 and CUDA 12.7 to accelerate NeRF optimization and dense point sampling.

For robotic experiments were executed using ROS 2 Humble running on an NVIDIA Jetson AGX Orin 64 GB as the robot control computer. The manipulator platform as Fanuc CRX-5iA (6-DoF). An Intel RealSense D405 RGB-D camera was used for scene capture. The end-effector was a custom RGIC gripper, and gripper actuation was limited to open-close motion with a closing width constrained to the berry size to avoid excessive contact during approach validation.

3. Results and discussions

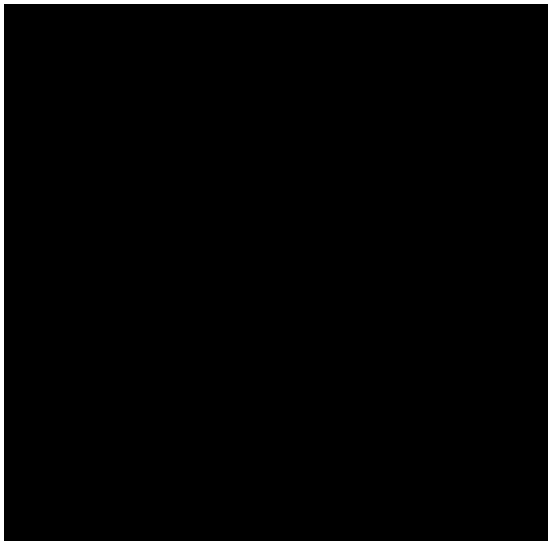
3.1. NeRF training convergence

We trained the NeRF model and evaluated intermediate checkpoints at 1 k, 5 k, 10 k, and 100 k iterations using SSIM and PSNR on a held-out test set. Table 2 summarizes reconstruction fidelity and training time.

The metrics indicate rapid early improvement followed by saturation. From 1 k to 10 k, SSIM increased by +0.041 and PSNR by +0.59 dB, reflecting visibly sharper geometry and fewer rendering artifacts. Beyond 10 k, gains were marginal: SSIM improved by only +0.009 at 100 k, while PSNR slightly decreased (18.54–18.46 dB). Importantly,



(a) Distribution of pre-grasp pose without berry-to-stem orientation constraints



(b) Distribution of pre-grasp pose derived from berry-to-stem orientation constraints

Fig. 8. Comparison of pre-grasp pose distributions for a representative bunch. (a) Without orientation constraints: pre-grasp frames cluster near a default global orientation, concentrating approach directions over a narrow angular range. (b) With stem-aligned orientation constraints: pre-grasp frames are spatially distributed to match local branch geometry, producing the directional diversity necessary for collision-free insertion.

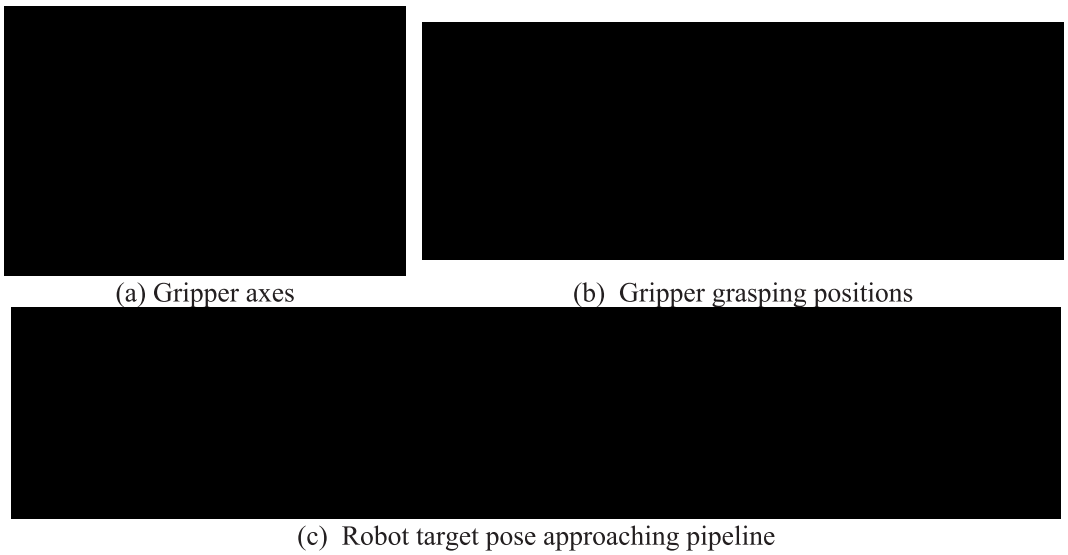


Fig. 9. Robot manipulation framework. (a) Gripper coordinate frame with approach axis (blue) aligned to berry poses. (b) Pre-grasp pose distribution (RGB frames) for all detected berries, offset 80 mm along approach directions. (c) Motion execution sequence: initial pose, collision scene setup, pre-grasp approach, and target insertion.

Table 2
NeRF reconstruction quality and training computational cost.

Iterations	SSIM ↑	PSNR ↑	Training Time (mins)
1 k	0.509	17.95	1:30
5 k	0.542	18.45	2:47
10 k	0.550	18.54	3:56
100 k	0.559	18.46	30:52

the training time increased by approximately 7.8 times from 10 k to 100 k. Collectively, these results suggest that 5 k to 10 k iterations provide the best trade-off between reconstruction quality and computational cost for downstream point cloud extraction pipeline.

Fig. 10 qualitatively compares the rendered test view (top row) and the exported point cloud (bottom row) across iteration counts. At very low iteration count (100), both rendered image and exported geometry are unstable, producing blurred appearance and diffuse, low-density point cloud with substantial floating noise. By 1 k iterations, the model begins to recover the coarse silhouette of the bunch, but berry boundaries remain poorly separated and the point cloud still contains many invalid points, particularly around high-frequency regions.

At 5 k iterations, the reconstructed surface becomes substantially more coherent: individual berries start to appear as separable in the point cloud, and the rendered view shows reduced blur and improved edge definition. At 10 k iterations, geometry is most consistent for downstream processing: berry structure is clearer, outlier density is reduced, and the reconstructed bunch exhibits the most stable berry-to-stem continuity among the tested checkpoints, which is critical for reliable sphere fitting and collision-aware manipulation. Although 10 k iterations produced slightly higher SSIM and PSNR, 5 k was selected as

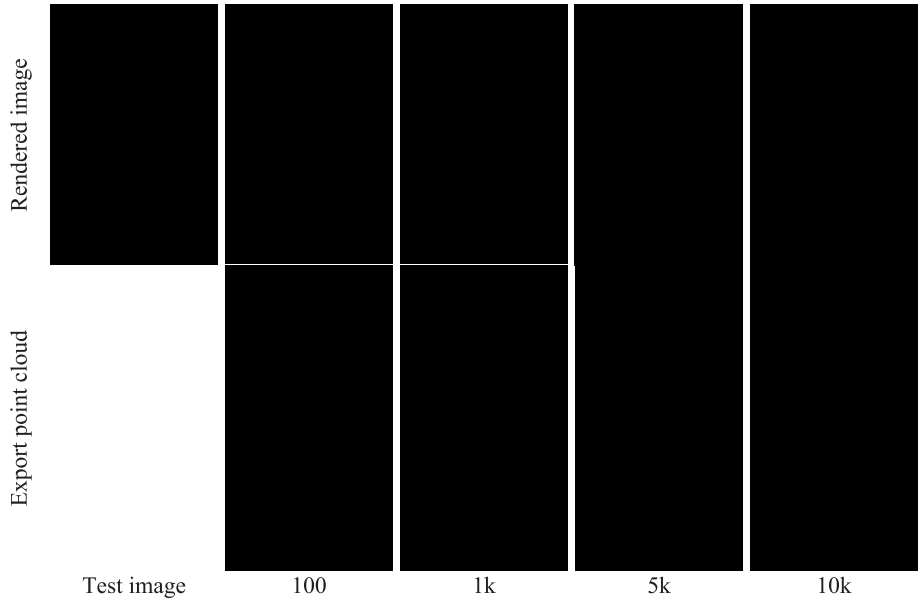


Fig. 10. Qualitative comparison of NeRF training convergence for grape bunch reconstruction. Top row: reference RGB image and NeRF renderings at increasing training iterations (100, 1 k, 5 k, and 10 k). Bottom row: corresponding point clouds extracted via ray-based depth back-projection, showing progressive improvement in geometric completeness and reduced background noise.

the default because it achieved comparable reconstruction quality with lower computational cost.

3.2. View-sampling requirement for NeRF reconstruction and berry detection

To clarify the inference-stage input requirement, we conducted a view-sampling ablation study using the full 360° acquisition sequence as the reference. The original sequence contained 279 posed RGB images captured at approximately 1.29° angular increments. We generated reduced training sets by uniformly subsampling the original trajectory at angular intervals of 3°, 9°, 12°, 20°, and 25°, and trained a separate NeRF model for each setting using the same reconstruction and point-cloud export parameters.

Fig. 11 shows the sampled camera viewpoints for each angular interval. As the angular interval increases, the number of training views decreases and the overlap between neighboring views become smaller. The 25° settings therefore represent the sparest sampling condition, while the 3° settings represent the densest reduced-view condition. This experiment evaluates the trade-off between acquisition density reconstruction fidelity, and downstream RANSAC-based berry detection.

For each sampling condition, we compared the reconstructed point cloud with the full-view 279-image reconstruction. Let A denote the point cloud reconstructed from sampled views and B denote the point cloud reconstructed from the full 279-view sequence. We computed bidirectional nearest-neighbor distance between A and B. The $A \rightarrow B$ direction measures how closely the sampled reconstruction agrees with the full-view reference, whereas the $B \rightarrow A$ direction measures how completely the sampled reconstruction preserves geometry present in the full-view reconstruction. For each direction, we report the mean and 95th percentile of the nearest-neighbor distances as robust geometric discrepancy metrics. In addition, we report the symmetric Hausdorff maximum, $H_{max} = max(maxd_{A \rightarrow B}, maxd_{B \rightarrow A})$, as a conservative worst-case deviation. Because H_{max} is sensitive to isolated floating points, it is interpreted together with the mean and 95th percentile distances rather than as the sole reconstruction-quality indicator.

Fig. 12 shows rendered images and point cloud distance maps for each sampling condition. Quantitative results are summarized in Table 3. The geometric distance increased as the angular sampling became sparser. At 3° sampling, the sampled reconstruction was very close to the full-view reference, with mean distances of 0.286 mm for A → B and 0.281 mm for B → A. At 25° sampling, the mean distances increased to 0.801 mm and 0.824 mm, respectively. The 95th-percentile distance showed the same trend, increasing from 0.697/0.667 mm at 3° to 2.338/1.9262 mm at 25°. This confirms that denser view sampling produces point clouds that are geometrically closer to the full 279-view reconstruction, while the sparsest 25° condition produces the largest deviation.

Downstream berry detection followed the same trend. The ground-truth berry counts for this bunch was 55. The 25° and 20° settings detected only 45 and 48 berries, respectively, indicating that sparse angular sampling loses local berry surface structure needed for reliable RANSAC sphere fitting. The 12° settings improved detection to 53 berries, while the 9° and 3° settings detected all 55 berries. Rendering metrics were also highest at 9° and 3° sampling, with the 9° settings achieving PSNR of 18.33 dB and SSIM of 0.61, and the 3° setting achieving PSNR of 18.0 dB and SSIM of 0.59.

Overall, this ablation shows that the proposed method requires a posed multi-view image sequence rather than a single image. In this representative case, 9° and 3° angular sampling detected all 55 berries, while 12°, 20°, and 25° sampling progressively lost berries. Thus, approximately 9° angular spacing can be interpreted as an empirical lower bound for this bunch. Nevertheless, the main pipeline uses the full 279-view sequence because the objective of robotic berry thinning is not only correct counting, but robust reconstruction of berry surfaces, occluded regions, and fine stem geometry for downstream segmentation and manipulation planning.

3.3. Berry detection and localization from robot-derived RealSense D405 reconstruction

3.3.1. Evaluation metrics

Berry detection performance was evaluated at the bunch level and berry-center level. For each bunch i, let N_i denote the manual ground-truth berry count and $\widehat{N}_i$ denote the number of berries detected by RANSAC sphere fitting. The absolute counting error is defined as $|\widehat{N}_i - N_i|$. Across B evaluated bunches, counting performance was summarized using mean absolute error (MAE) and root mean square error (RMSE):

$$MAE = \frac{1}{B}\sum_{i=1}^{B}|\widehat{N}_i - N_i| \tag{10}$$

$$RMSE = \sqrt{\frac{1}{B}\sum_{i=1}^{B}(\widehat{N}_i - N_i)^2} \tag{11}$$

To evaluate berry-center localization accuracy, detected berries were first matched to ground-truth berries using 3D IoU-based matching. Each detected berry and ground-truth berry was represented as a sphere parameterized by its center and radius. A detected berry was considered matched to a ground-truth berry when their IoU exceeded a threshold τ_{IoU}.

For two spherical berry instances B_i and $\widehat{B}_j$, 3D IoU as defined as the ratio between their volumetric intersection and volumetric union:

$$3DIoU(B_i, \widehat{B}_j) = \frac{Vol|B_i \cap \widehat{B}_j|}{Vol|B_i \cup \widehat{B}_j|} \tag{12}$$

A detected berry was considered matched to a ground-truth berry when $3DIoU(B_i, \widehat{B}_j) \geq \tau_{IoU}$. In this evaluation, $\tau_{IoU} = 0.25$ was used.

One-to-one matching was enforced so that each detected berry could

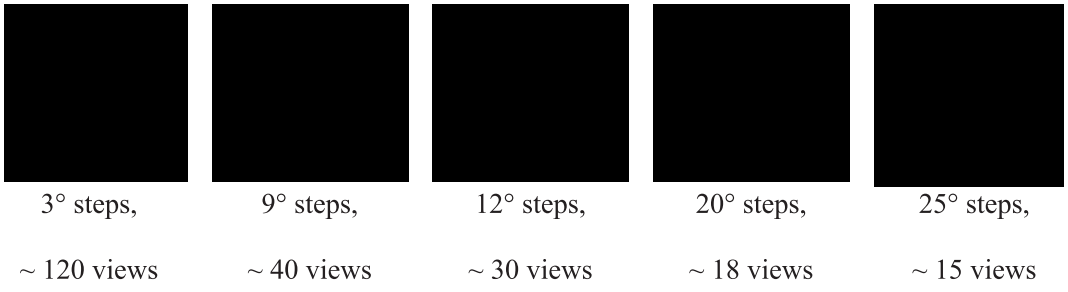


Fig. 11. View-sampling strategy for evaluating inference-stage input requirements. The original 360° acquisition sequence contained 279 posed RGB images. Reduced training sets were generated by selecting views at angular intervals of 3°, 9°, 12°, 20°, and 25°. Each circle shows the selected camera viewpoints around the grape bunch.

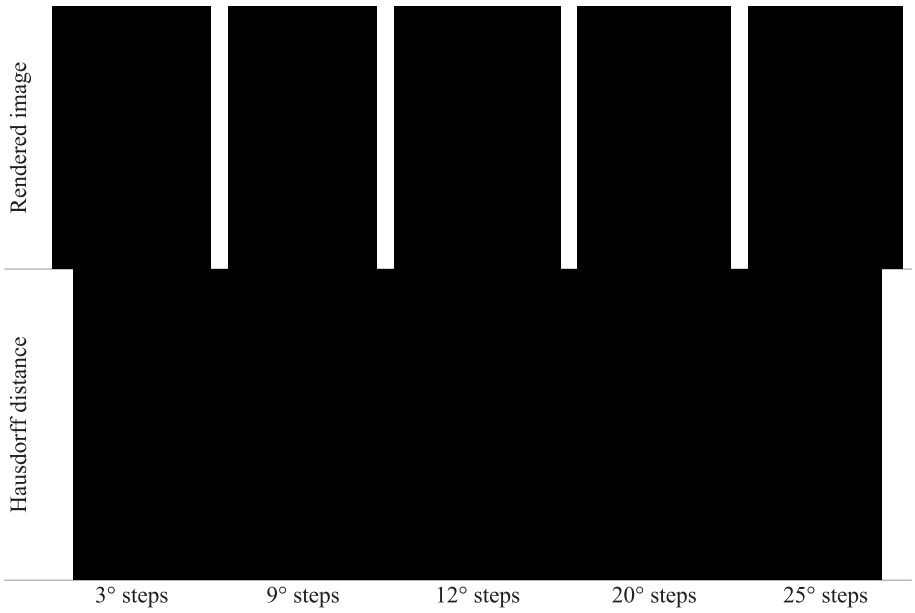


Fig. 12. Effect of angular view sampling on rendered appearance and point-cloud geometry. Top row: rendered views from NeRF models trained with 3°, 9°, 12°, 20°, and 25° angular sampling. Bottom row: point-wise $A \rightarrow B$ nearest-neighbor distance maps, where A is the sampled-view reconstruction and B is the full 279-view reference reconstruction. Denser sampling preserves the bunch geometry more completely, whereas sparse 20° and 25° sampling produces missing local berry surfaces and increased geometric deviation.

Table 3
View-sampling ablation results.

Angular step	PSNR	SSIM	$A \rightarrow B$ mean (mm)	$B \rightarrow A$ mean (mm)	$A \rightarrow B$ 95th (mm)	$B \rightarrow A$ 95th (mm)	Detected / Ground truth	Count error	H_{max}(m)
3°	18.0	0.59	0.286	0.281	0.697	0.667	55/55	0	0.07
9°	18.33	0.61	0.304	0.308	0.752	0.717	55/55	0	0.08
12°	16.85	0.47	0.372	0.359	0.952	0.856	53/55	2	0.12
20°	16.89	0.46	0.587	0.515	1.631	1.191	48/55	7	0.43
25°	17.08	0.52	0.801	0.824	2.338	1.962	45/55	10	0.50

match at most one ground-truth berry, and each ground-truth berry could be assigned to at most one detection. For each matched pair j, let $\hat{c}_j$ and c_j denote the detected and ground-truth berry centers, respectively. The mean center distance error was defined as:

$$\bar{e}_c = \frac{1}{M} \sum_{j=1}^{M} \left| \hat{c}_j - c_j \right|_2 \tag{13}$$

3.3.2. Counting and localization accuracy

We first evaluated berry detection using the robot-derived D405 reconstruction, corresponding to the eye-in-hand acquisition pipeline described in Section 2.2.1. This evaluation is the primary robot-integrated segmentation result because the D405 RGB images were captured from the robot-mounted sensor and the camera poses were obtained from the robot kinematic chain through ROS 2 TF2. Therefore, the reconstructed point cloud and detected berry centers were expressed directly in the robot base frame and could be used for downstream stem reconstruction, 6-DoF target pose estimation, and robot motion planning.

Table 4 summarizes the counting and localization results for 10 grape bunches. RANSAC sphere fitting achieved an MAE of 0.50 berries and an RMSE of 0.71 berries. The absolute counting error was at most one berry for all tested bunches, indicating that the robot-mounted D405 reconstruction preserved sufficient local berry surface geometry for reliable sphere fitting. The center distance error was 2.712 mm on average, confirming that the detected berry centers were geometrically consistent with manual ground-truth annotations.

Table 4
Berry detection accuracy using robot-derived D405 reconstruction. Manual berry counts are used as ground truth. D405 detected counts were obtained by applying RANSAC sphere fitting to NeRF-reconstructed point clouds generated from robot-mounted D405 RGB images and TF2-derived camera poses. Count error denotes absolute counting error. Mean center distance error is computed over matched berry pairs.

Grape ID	Ground-truth count	Detect count	Count error	$\bar{e}_c$(mm)
ID 1	44	45	1	4.32
ID 2	41	41	0	2.14
ID 3	43	43	0	3.60
ID 4	37	37	0	3.54
ID 5	54	54	0	2.08
ID 6	39	40	1	2.52
ID 7	50	51	1	1.89
ID 8	55	56	1	3.23
ID 9	45	46	1	2.55
ID 10	62	62	0	1.25
MAE			0.50	2.712
RMSE			0.71	2.854

3.3.3. Qualitative segmentation results from robot-mounted reconstruction

Fig. 13 shows representative RealSense D405 reconstructions and corresponding RANSAC sphere fitting results. Despite the relatively compact RGB sensor, individual berry surfaces were sufficiently reconstructed for geometric primitive fitting. Most berries were detected as distinct spherical primitives, while small errors occurred mainly around peripheral berries and partially occluded regions where the reconstructed surface support was limited. These qualitative results are

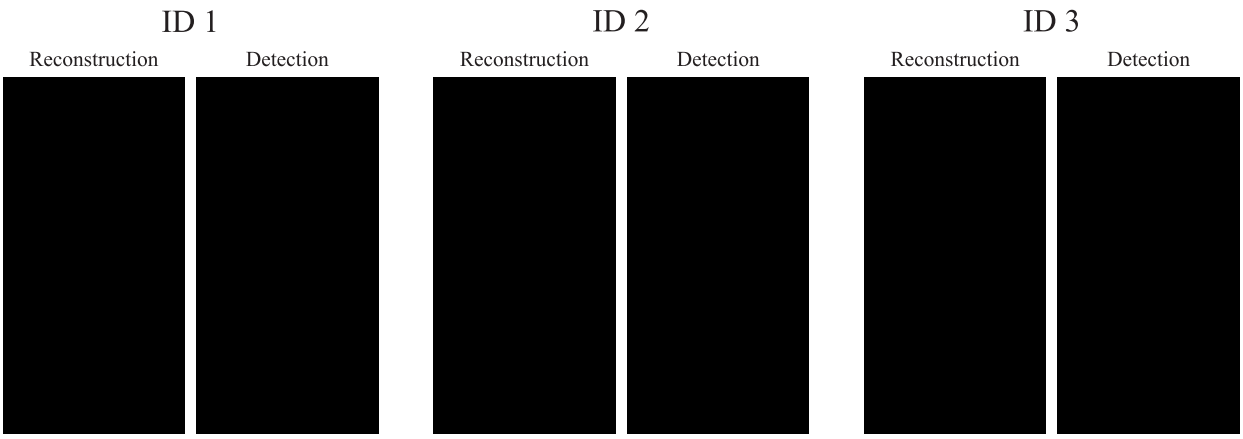


Fig. 13. Qualitative berry detection results from robot-derived-pose D405 reconstruction. Representative NeRF-reconstructed point clouds and corresponding RANSAC sphere fitting results from the robot-mounted D405 acquisition pipeline are shown. Detected berry spheres are colored by instance. The results show that D405 RGB reconstruction with TF2-derived poses preserves sufficient berry geometry for instance-level detection and downstream robot planning.

consistent with the low counting error reported in Table 4 and support the use of the RealSense D405 as the primary robot-mounted sensing configuration.

3.4. Comparison of robot-mounted iPhone and RealSense acquisition for reconstruction and berry segmentation

To examine whether using a higher-quality RGB sensor improves reconstruction and berry detection, we conducted a paired robot-mounted acquisition comparison between the D405 RGB image and an iPhone RGB image. Both sensors were mounted on the Fanuc CRX-5iA end-effector and used to capture the same 10 grape bunches under comparable acquisition conditions. The D405 sequences used camera poses obtained from the robot kinematic chain through ROS 2 TF2 as described in Section 2.2, whereas the iPhone sequences used Record3D poses. Fig. 14 shows the robot-mounted iPhone acquisition setup used for the paired sensor comparison.

This comparison is not intended to isolate RGB image resolution alone, because the two acquisition pipelines differ in optical center, intrinsic parameters, focus behavior, exposure characteristics, and pose-estimation source. Instead, the experiment compares two practical robot-mounted RGB acquisition pipelines: a compact D405 setup with robot-derived poses and a smartphone-based RGB setup with higher nominal image quality.

Table 5 compares berry counting accuracy between the D405 and iPhone acquisition pipelines. Across 10 bunches, the D405 pipeline achieved an MAE of 0.50 berries and an RMSE of 0.71 berries, whereas the iPhone pipeline achieved an MAE of 1.70 berries and an RMSE of 2.02 berries. The D405 result was better in 8 of the 10 bunches and tied in the remaining 2. The iPhone detections showed a systematic under-counting tendency, while the D405 detections produced only small count deviations. This result does not imply that the D405 RGB sensor has higher intrinsic image quality; rather, it suggests that stable focus, synchronized robot-derived poses, and consistent acquisition can be more important for downstream RANSAC counting than nominal RGB

Table 5
Quantitative comparison of berry counting accuracy between robot-mounted iPhone and RealSense D405 acquisition pipelines. Manual berry counts are used as ground truth, and absolute counting errors are reported for RANSAC sphere fitting results from NeRF-reconstructed point clouds.

Grape ID	Ground-truth count	iPhone count	RealSense count	iPhone error	RealSense error
ID 1	44	40	45	4	1
ID 2	41	42	41	1	0
ID 3	43	44	43	1	0
ID 4	37	35	37	2	0
ID 5	54	54	54	0	0
ID 6	39	38	40	1	1
ID 7	50	48	51	2	1
ID 8	55	52	56	3	1
ID 9	45	43	46	2	1
ID 10	62	61	62	1	0
MAE				1.70	0.50
RMSE				2.02	0.71



Fig. 14. Robot-mounted iPhone acquisition setup used for paired sensor comparison. (a) iPhone RGB camera mounted near the robot end-effector. (b) Example iPhone acquisition trajectory and reconstructed camera pose around the grape bunch.

resolution alone under the tested robot-mounted acquisition conditions.

Fig. 15 visualizes the same trend by plotting iPhone detections against RealSense D405 detections for all 10 bunches. Most points lie above the diagonal line, indicating that the D405 pipeline detected more berries than the iPhone pipeline for the same bunches.

Fig. 16 shows an example failure case from the robot-mounted iPhone acquisition. Although the iPhone provided higher-resolution RGB images, several sequences contained out-of-focus frames. These frames reduced multi-view photometric consistency during NeRF optimization and produced incomplete point clouds with missing berry surfaces. Because RANSAC sphere fitting depends on sufficient local surface support, these incomplete surfaces caused missed detections and contributed to the under-counting observed in Table 5.

Fig. 17 further compares paired D405 and iPhone reconstructions across representative bunches. The iPhone reconstructions are visually smooth in some regions, but several contain missing local berry surfaces, particularly around peripheral or partially occluded berries. In contrast, the D405 reconstructions are sometimes less smooth but preserve more consistent berry-level surface support, which explains their lower counting error. Together, Figs. 16 and 17 indicate that reconstruction completeness and acquisition stability are more important for RANSAC-based berry counting than nominal RGB resolution alone.

3.5. Large-scale high-resolution benchmark for berry instance segmentation

3.5.1. Dataset overview

After validating the proposed berry detection pipeline using robot-derived-pose D405 reconstruction in Section 3.3, we further evaluated the RANSAC sphere fitting method on a larger high-resolution benchmark dataset. This dataset comprised 52 grape bunches captured during March-April 2024 using an iPhone 15 at 3840 × 2160 pixels and 60 fps. Each bunch was recorded as a 360° handheld video, from which 80–120 frames were extracted using FFmpeg to provide sufficient view overlap for multi-view reconstruction (in Fig. 18).

This dataset was used only for large-scale berry instance segmentation and counting evaluation. It was not used for robot control because camera poses were estimated using COLMAP's structure-from-motion pipeline. NeRF models were trained following Section 2.1 methodology. Since COLMAP produces scale-ambiguous reconstructions, metric scale was recovered using a reference calibration tag (45.0 mm height) positioned adjacent to each bunch during capture (in Fig. 26). The reconstructed tag height determined a per-bunch scale factor applied uniformly to all coordinates.

Berry diameters range 10.2–24.8 mm (mean: 17.43 mm) as shown in Fig. 19a with bunch sizes ranging from 12 to 89 berries (mean: 53.5 ± 18.2), illustrated per sample in Fig. 19b.

3.5.2. Instance-level segmentation metrics

Instance-Instance-level berry segmentation was evaluated using the 3D IoU defined in Section 3.3.1. For both RANSAC and SoftGroup++, each predicted berry instance was represented as a parameterized sphere over the evaluated point cloud. For SoftGroup++, the mask corresponds to the predicted instance output. For RANSAC, the mask corresponds to the set of points assigned to each fitted sphere.

Predicted and ground-truth berry instances were matched using one-to-one IoU-based matching. A prediction was counted as a true positive when its point-mask IoU with an unmatched ground-truth instance exceeded the evaluation threshold. Unmatched predictions were counted as false positives, and unmatched ground-truth instances were counted as false negatives. Average precision and recall were reported at IoU thresholds of 0.25 and 0.50, denoted as AP@0.25, AP@0.50, AR@0.25, and AR@0.50. Because berry was the only evaluated object class, all reported metrics correspond to single-class berry instance segmentation performance.

For matched true-positive instances, berry-center localization error was computed using the same definition as in Section 3.3.1.

3.5.3. Berry segmentation performance

Using the evaluation metrices defined in the previous section, the proposed pipeline achieves high and consistent performance across both datasets. Precision and recall exceed 0.93 on average. The average center localization error remains within a few millimeters, and radius estimation errors fall within plausible ranges for grape berries.

Table 6 compares RANSAC sphere detection (Section 2.3) against SoftGroup++ (Vu et al., 2022), a learning-based point cloud instance segmentation method. Results are averaged over all 52 bunches. RANSAC achieves 92.1% average precision and 94.5% recall, substantially outperforming SoftGroup++ (82.8% AP, 88.4% AR). The performance advantage is consistent across IoU thresholds, with RANSAC

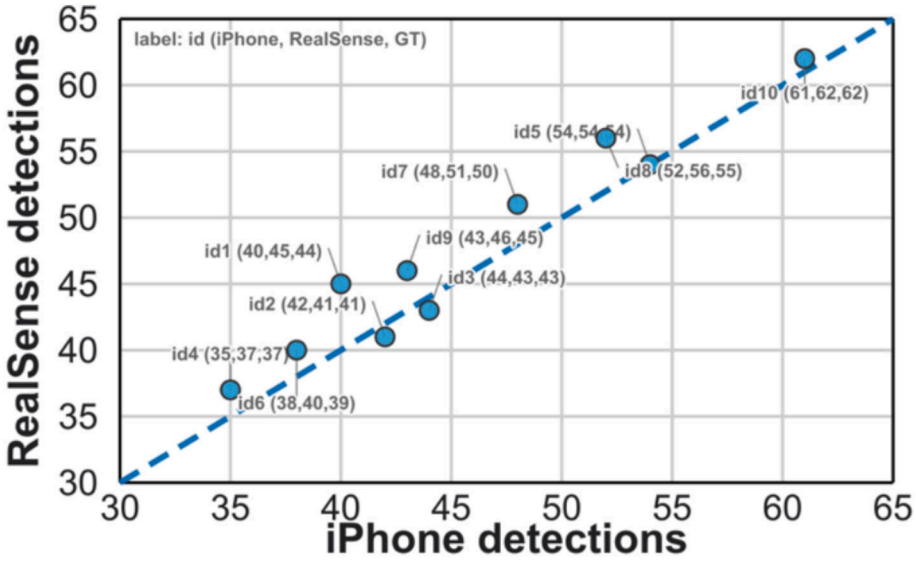


Fig. 15. Comparison of berry counts obtained from robot-mounted iPhone and RealSense D405 reconstructions across 10 grape bunches. The x-axis shows iPhone-based detections, and the y-axis shows RealSense D405-based detections. Each point corresponds to one bunch and is labeled as (iPhone, RealSense, GT), where GT denotes the manual count. The dashed diagonal indicates equal counts between the two sensing pipelines. Points above the diagonal indicate cases where the RealSense pipeline detected more berries than the iPhone pipeline.

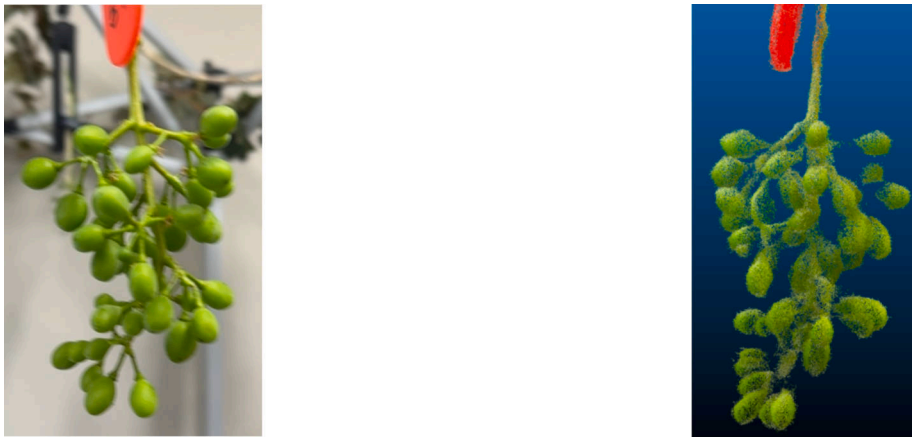


Fig. 16. Example of out-of-focus iPhone frames causing incomplete reconstruction. Defocused image frames reduce multi-view photometric consistency and lead to incomplete NeRF point clouds, which can cause missed RANSAC sphere detections.

maintaining 88.2% AP even at IoU = 0.50. while SoftGroup++ drops to 83.0%. Importantly, RANSAC exhibits superior localization accuracy (e_c = 0.07 cm). The superior performance of RANSAC shape detection can be attributed to three factors: (1) strong geometric priors matching actual berry, (2) robustness to incomplete point clouds due to parametric fitting, and (3) deterministic behavior eliminating training data dependency. Fig. 20 shows failure cases of SoftGroup++ in larger berry sizes where RANSAC does not failed to detect berries.

Beyond individual berry detection, accurate bunch-level counting is critical for yield estimation and thinning decision support. Table 7 compares our approach against recent grape counting methods. RANSAC-based detection achieves MAE of 1.1 berries and RMSE of 1.7 berries, representing 52% error reduction compared to the best prior 2D method (Woo et al., 2023: MAE 2.3) and 71% reduction compared to RGB-based approaches (Buayai et al., 2021: MAE 3.79).

Fig. 20 provides qualitative comparison of RANSAC and SoftGroup++ segmentation results across representative bunches spanning the difficulty (sparse, moderate, dense packing).

3.6. Bunch structure reconstruction

Direct ground-truth evaluation of grape bunch skeletonization is not feasible, as no physical reference centerline can be reliably obtained, and manual annotation of branch centerlines in occluded point clouds introduces unacceptable subjectively. We therefore adopt a two-level evaluation strategy applied to the same subset of 8 bunches comprising 491 berries: first, an indirect geometric consistency evaluation that quantifies how accurately the extracted skeleton captures the observed stem point cloud geometry; second, a topological assignment evaluation that assesses how reliably each berry is assigned to its current parent stem segment.

Geometric consistency evaluation. The evaluation pipeline is illustrated in Fig. 21. The residual point cloud obtained after removing RANSAC berry inliers (Fig. 21a) was first contracted into a 1D skeleton via Laplacian-based contraction (LBC), then connected into a tree topology via minimum spanning tree (MST) construction (Fig. 21b and c). Each MST edge was replaced by a fitted cylinder with fixed radius. The resulting cylinder surfaces were densely sampled (Fig. 21d) to generate a synthetic point cloud $P_{stem}^{'}$ that is geometrically comparable to the original residual cloud P_{stem}.

Geometric fidelity is evaluated using directional Hausdorff distances between the synthetic cylinder point cloud $P_{stem}^{'}$ (Fig. 22a) and the original residual point cloud P_{stem} (Fig. 22b). The cylinder fit direction ($P_{stem}^{'} \rightarrow P_{stem}$) measures how closely fitted cylinders approximate observed stem geometry, while the branch coverage direction ($P_{stem} \rightarrow P_{stem}^{'}$) measures how completely the cylinder model captures the original stem structure. Results are summarized in Table 8.

Both directions achieve sub-millimeter mean accuracy, confirming that the Laplcian-based contraction produces geometrically faithful skeleton representations. The per-point distance map (Fig. 22c) shows that errors are spatially concentrated at fine secondary pedicels and peripheral branch tips, while the main rachis and primary lateral branches exhibit close agreement (predominantly blue, < 1 mm). The elevated $P_{stem} \rightarrow P_{stem}^{'}$ maximum of 15.17 mm reflects isolated MST edges bridging spatial gaps at branching junctions; this affects fewer than 1% of points (99th percentile: 2.08 mm) and does not propagate to berry orientation estimates, which depend on the direction to the nearest skeleton point rather than global surface accuracy.

Topological assignment evaluation. Geometric accuracy alone does not guarantee that each berry is assigned to its anatomically correct parent stem. To evaluate topological correctness, we visualize berry-to-stem correspondences by colorizing each berry's point cloud according to its assigned stem segment, as shown in Fig. 23a. This visualization enables human inspection of assignment quality across the full bunch without requiring explicit ground-truth labelling.

However, topological accuracy cannot be assessed from geometric distance alone, as two distinct failure modes produce incorrect assignments despite geometrically plausible skeleton representations. These are illustrated in Fig. 23b and c.

The first failure mode, shown in Fig. 23b, arises from biological ambiguity: berries from adjacent sub-branches physically penetrate neighboring clusters, causing a berry's nearest skeleton point to belong to an adjacent branch rather than its true anatomical parent. The orange-highlighted berries indicate shared assignments to stem.

The second failure mode, shown in Fig. 23c, arises from insufficient reconstruction resolution: fine sub-branches that is insufficient reconstruction resolution are absent from the skeleton, causing the berry to be assigned to a geometrically closer but anatomically incorrect stem segment. In Fig. 23c contrasts the actual stem attachment direction with the incorrectly assigned closest reconstructed segment.

Both failure cases produce shared berry-to-stem assignments where a single stem segment is matched to two or more berries but arise from fundamentally different causes. To quantify this, a shared-assignment rate error was defined as a stem segment matched to two or more berries across the same 8-bunch evaluation subset. The mean error rate was 14.1 ± 9.3% (range: 4.3–28.2%), with two bunches exceeding 25%. Most shared assignments involved pairs of berries mapped to the same stem, with no triple assignments observed. A moderate positive correlation between bunch size and the absolute number of shared berries

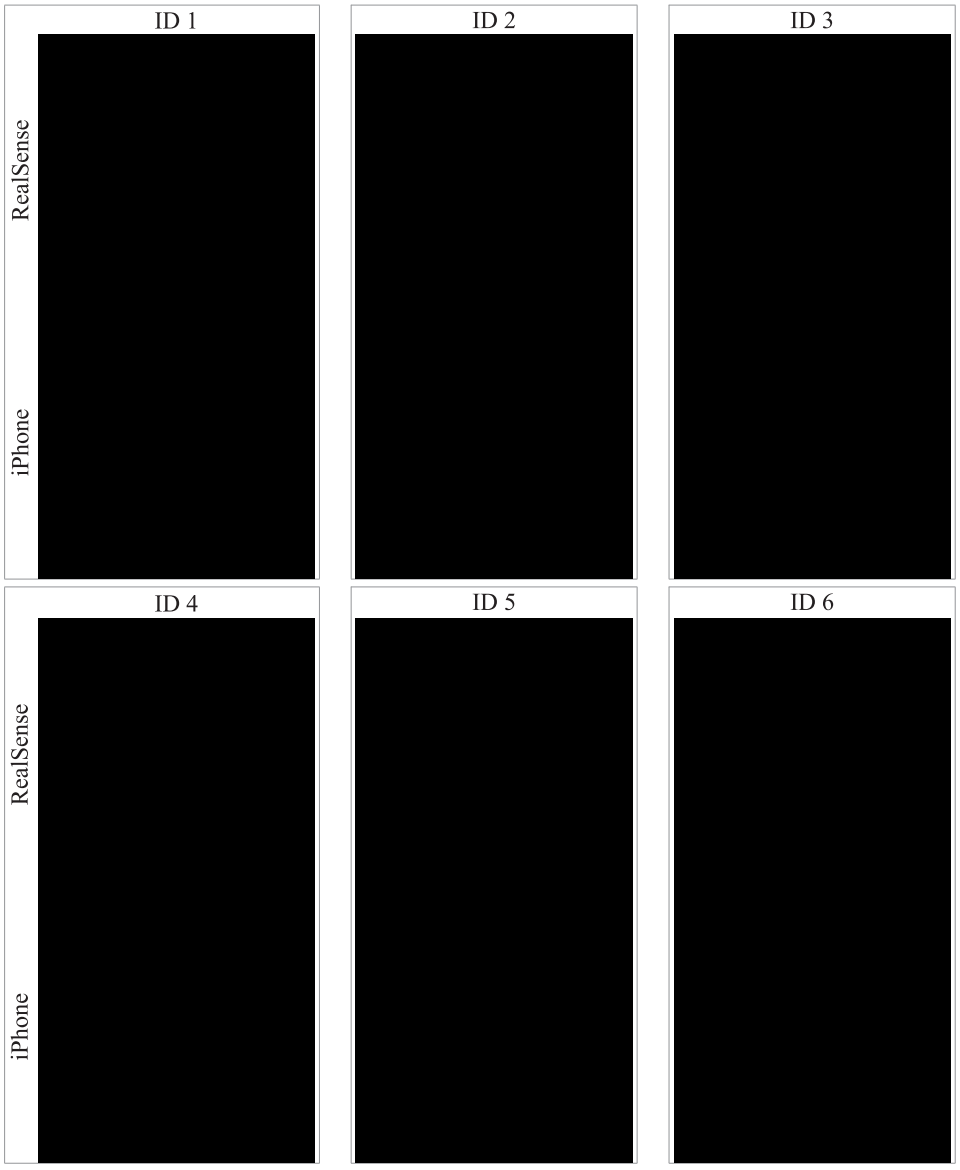


Fig. 17. Qualitative comparison of RealSense D405 and iPhone acquisition pipelines. For each representative bunch, the NeRF-reconstructed point cloud and corresponding RANSAC sphere fitting result is shown. Several iPhone reconstructions contain incomplete berry surfaces caused by out-of-focus frames, resulting in missed detections. In contrast, the D405 reconstructions are generally more complete and produce more consistent berry detections under the tested robot-mounted acquisition conditions.

($R^2 = 0.63$) indicates that larger, denser bunches accumulate more topological ambiguity, consistent with the increased physical overlap between sub-branches at higher packing densities.

3.6.1. Berry orientation optimization

To establish manipulation-ready target poses, we optimize berry orientation to align the end-effector approach axis with the estimated stem attachment direction (Section 2.4.2). We compared five rotation parameterizations for the gradient-based optimization: axis-angle, quaternion, Euler angles, 5D representation, and the 6D representation of Zhou et al. (2019).

Table 9 summarizes final angular errors after 2000 optimization iterations, averaged across all detected berries in single grape bunch. Fig. 24 shows average error curves for each rotation representation. The 6D and quaternion representations achieve lower final error (0.09° and 0.10°, respectively).

Fig. 25 provides a qualitative comparison by visualizing the optimized end-effector frames at each berry target. Axis-angle optimization (Fig. 25a) produces inconsistent orientations (blue axis), exhibit more than 40° deviation in approach direction, could potentially create collision risk during sequential manipulation. The 6D representation (Fig. 25b) yields more consistent orientation outcome where the blue axis is pointing toward local stem point (red dot), whereas axis angle representation shows greater variation in the optimized orientations across berries.

3.7. Robot manipulation

3.7.1. Metric scale verification

In our system, camera poses are provided in metric coordinate via the robot TF tree, and the NeRF scene is trained using these metric extrinsic. As a result, the exported NeRF point cloud preserves metric scale and is directly compatible with downstream geometric processing and robot motion planning.

To validate that metric scale is preserved through the pipeline, we reconstructed a reference object (Fig. 26a) has known height dimensions



Fig. 18. NeRF reconstruction diversity across dataset. Four representative bunches demonstrating reconstruction quality across varying bunch density (sparse to dense), berry size (10–20 mm), and development stage (early to mature).

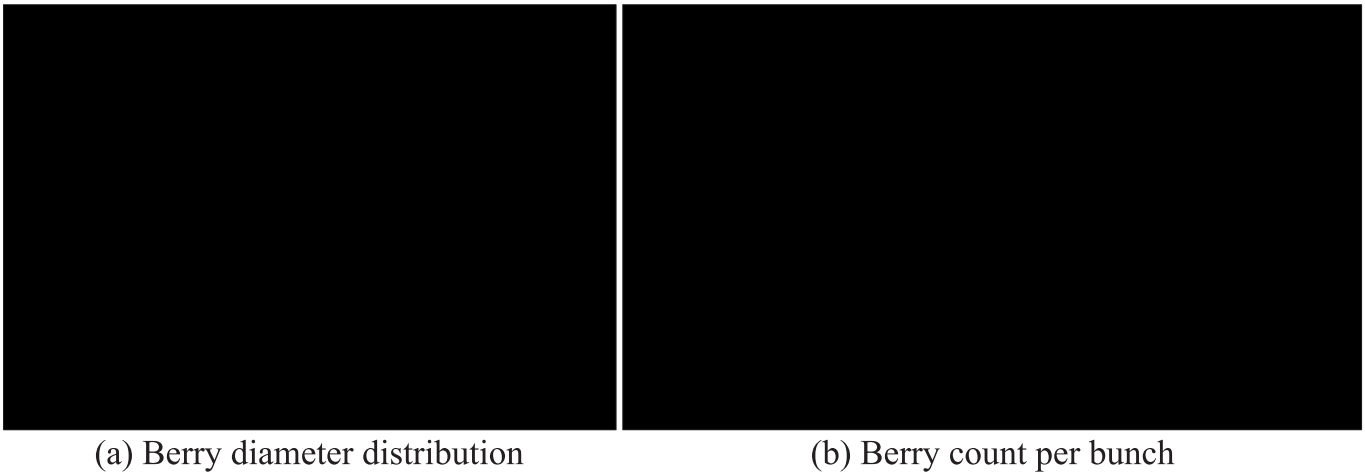


Fig. 19. Distribution of berry size and berry count in the annotated benchmark dataset. (a) Berry diameter distribution estimated from per-berry instance point clouds. Dashed vertical lines indicate the mean diameter of 17.43 mm and median diameter of 16.39 mm. (b) Number of annotated berries per bunch. The dashed horizontal line indicates the dataset mean of 53.5 berries per bunch, highlighting inter-bunch variation in cluster density and segmentation difficulty.

Table 6
Quantitative comparison of berry instance detection performance (n = 52 bunches). Results compare RANSAC shape detection against SoftGroup++. AP and AR denote average precision and recall, respectively, with subscripts indicating IoU thresholds (0.25, 0.50).

Method	**AP**	**AP**@0.25	**AP**@0.50	**AR**	**AR**@0.25	**AR**@0.50	$\overline{e_c}$(cm)
RANSAC	**0.914**	**0.921**	**0.882**	**0.945**	**0.941**	**0.910**	**0.07**
SoftGroup++	0.828	0.839	0.830	0.884	0.890	0.873	0.18

of 45 mm. The corresponding dimension measured in point cloud (Fig. 26b) is 45.9 mm, confirming metric scale preservation without requiring additional calibration objects or scaling procedures. This validation establishes that reconstructed berry positions can be used directly for motion planning in the robot base frame.

3.7.2. *Manipulator trails*

Experimental setup. We evaluate end-to-end manipulation feasibility through 35 sequential approaching trials using the complete perception-to-motion pipeline described in Section 2. Each trial commanded the robot to execute a two-stage motion primitive: (i) global collision-free motion from the current configuration to a pre-grasp pose aligned with the detected berry's estimated approach vector (Fig. 27b), followed by (ii) an 80 mm straight-line Cartesian insertion to the final berry target pose (Fig. 27c). A trial was marked as a pre-grasp success if the robot reached the pre-grasp pose without collision, and as a target success if it subsequently completed the full insertion to the berry surface. Elapsed time was recorded at each stage. A single failure did not terminate the sequence; the robot returned to the safe home configuration before proceeding to the next target.

Success criteria. We report three outcome metrics: (1) pre-grasp reachability, whether the robot reached the pre-grasp pose from its

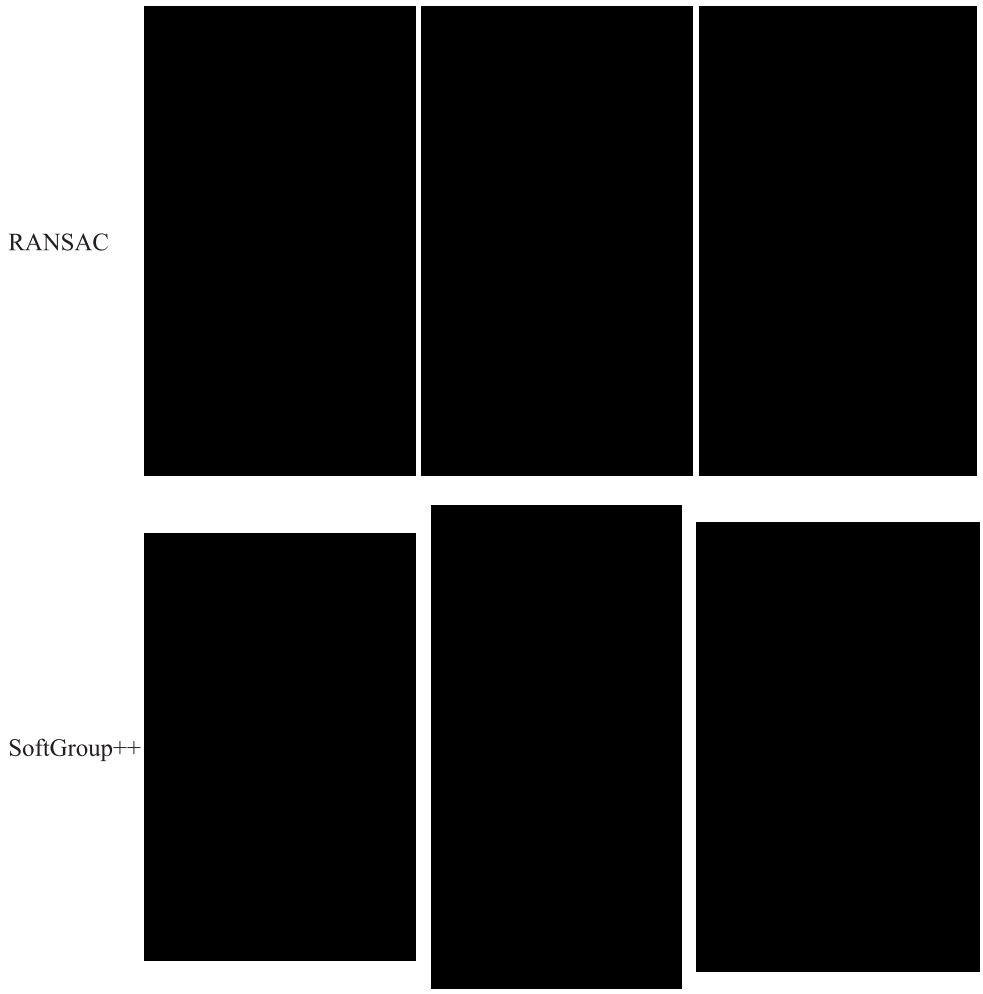


Fig. 20. Qualitative comparison of berry detection methods on representative grape bunches. Top row: RANSAC sphere detection results with fitted spheres, colored by instance. Bottom row: SoftGroup++ instance segmentation with per-instance coloring. RANSAC maintains consistent detection quality across density levels and sizes, while SoftGroup++ exhibits systematic under-segmentation in dense regions (bottom-right).

Table 7
Bunch-level berry counting accuracy comparison. Prior methods operate on 2D RGB images; our method operates on NeRF-reconstructed 3D point clouds.

Previous	Approach	MAE	RMSE
Buayai et al. (2021)	2D detection + Machine learning	3.79	–
Woo et al. (2023)	2D detection + Machine learning	2.3	–
Du & Liu (2023)	2D multi-view fusion	–	4.72
Woo et al. (2023b)	Classical SfM + 3D projection	1.92	–
Our	**NeRF + 3D RANSAC Sphere Detection**	**1.1**	**1.7**

current configuration; (2) target reachability, whether the robot completed the full 80 mm linear insertion to the berry surface, conditioned on pre-grasp success; and (3) collision-free insertion, whether the final approach was completed without any collision contact with the bunch or support structure, as judged by the absence of force-torque anomalies and visual inspection. The conditional success rate is computed as target-reachable/pre-grasp-reachable, isolating IK feasibility at the constrained 6-DoF target pose from motion planning (in Fig. 28).

Main results. As depicted in Fig. 29a, the robot reached the pre-grasp pose for 30 of 35 targets (85.7%). Among these, 25 achieved successful final approach and collision-free insertion, yielding a conditional success rate of 83.3% (25/30). The five failures at the final approach stage resulted entirely from inverse kinematics infeasibility under the strict 6-DoF orientation constraint, where no valid joint configuration existed to satisfy the requested end-effector pose. No failures were attributed to collision avoidance or perception errors at this stage, confirming that the dominant bottleneck is IK feasibility rather than detection accuracy.

Ablation study: effect of approach orientation strategy. To isolate the contribution of structure-aware orientation estimation, we repeated the 35-trial sequence under two baseline strategies defined in Section 2.6: a radially-outward baseline, in which the approach direction for each berry is defined as the unit vector from the bunch centroid through the berry center, requiring no skeleton extraction; and our proposed stem-aligned strategy, in which approach directions are inferred from the reconstructed branch skeleton. Results are summarized in Table 10.

Stage-by-stage analysis. The two conditions exhibit a characteristic trade-off that manifests across the three stages, ultimately resolving in favor of stem-aligned orientation at the end-to-end level. At the pre-grasp stage, the unconstrained baseline achieves 100% reachability because the absence of an orientation constraint gives the IK solver maximum freedom to find joint configurations within the robot's workspace. In contrast, the stem-aligned condition imposes a specific 6-DoF target for the pre-grasp pose, reducing reachability to 85.7% as some orientations require wrist configurations near joint limits. This 14.3% reduction at the pre-grasp stage represents the cost of incorporating structural geometry into manipulation planning.

However, this cost is recovered at the target reachability stage.



Fig. 21. Skeleton-based stem reconstruction pipeline. (a) Residual point cloud after removing RANSAC berry inliers. (b) Laplacian-based contraction (LBC) producing a skeleton. (c) Minimum spanning tree (MST) connecting skeleton points into a tree topology. (d) Cylinder fitting along each MST edge, with dense surface sampling to generate the synthetic point cloud $\acute{P}_{stem}$. (e) Dense point cloud from fitted cylinder model surfaces (orange), showing spatial correspondence between reconstructed and observed stem geometry.



Fig. 22. Geometric evaluation of skeleton-based cylinder reconstruction via directional Hausdorff distance. (a) Synthetic point cloud sampled from fitted cylinder surfaces. (b) Original residual stem point cloud after berry removal. (c) Per-point distance map of the branch coverage direction (B → A), where each original stem point is colored by its distance to the nearest cylinder surface point. Blue indicates close agreement (<1 mm); red indicates regions where the original stem structure is not captured by the cylinder model, predominantly at fine secondary pedicels below reconstruction resolution.

Among targets where the pre-grasp was reached, the unconstrained baseline achieves only 65.7% target reachability (23/35 overall) because the unconstrained pre-grasp orientation frequently places the end-effector in a kinematically unfavorable posture for the subsequent constrained insertion. Conversely, the stem-aligned pre-grasp orientation is designed to be consistent with the final target pose, so once the pre-grasp is reached, the conditional probability of completing the insertion rises to 83.3% (25/30). The net effect is that target reachability

Table 8
Directional Hausdorff distance evaluation of skeleton-based cylinder reconstruction. All values in millimeters.

Metric	$\vec{P}_{stem} \rightarrow P_{stem}$ (Cylinder fit)	$P_{stem} \rightarrow \vec{P}_{stem}$ (Branch coverage)
Mean ± Std	0.66 ± 0.48	0.45 ± 0.31
Median	0.54	0.37
95th Percentile	1.51	1.08
99th Percentile	2.08	1.49
Max	15.17	2.21

is comparable across conditions (65.7% vs. 71.4%) despite the large difference in pre-grasp reachability.

The most significant difference between two methods occurs during the gripping stage. In the unconstrainted baseline, 5 of 23 target-reachable trails (21.7%) failed because the end-effector collide with neighboring berries during insertion. Because the approach vector was randomly oriented, it often missed the clearance space between berries, causing the gripper to push the target out of reach. Conversely, the stem-aligned approach follows the natural direction of the local branch, which almost entirely eliminated this failure mode. Only 1 of 25 target-reachable trails failed at gripping (4%), resulting in a 96.0% conditional grip success rate. Ultimately, the end-to-end grip success rate improved from 51.4% to 68.6%. This absolute gain of +17.2% highlights the value of structure-aware orientation estimation. Notably, this overall improvement occurred despite a 14.3% drop in pre-grasp reachability, providing that higher-quality approaches easily offset the limitations of stricter IK constraints.

Execution timing. Timing analysis of the stem-aligned trials is presented in Fig. 29b. Pre-grasp motion exhibited substantial variance (median: 6.0 s, IQR: 4.25–9.08 s), reflecting variable path length and complexity through the canopy structure depending on the robot's prior configuration and the spatial distribution of collision objects. In contrast, final approach timing was highly consistent (median: 2.0 s, IQR: 1.01–4.12 s), consistent with the deterministic nature of a fixed 80 mm straight-line Cartesian motion once a valid IK solution existed. The overall median time per successful berry approach was 8.5 s.

3.8. Runtime analysis and deployment implications

To evaluate practical applicability, we measured the processing time of the proposed scan-then-plan pipeline from image acquisition to robot-executable berry approach targets. Because the proposed method optimizes a NeRF model for each target bunch, inference in this study refers to the complete processing required to processed captured posed multi-view RGB sequence into berry-level geometric decisions, including berry segmentation, stem reconstruction, stem-aligned 6-DoF pose estimation, and motion execution targets. Table 11 summarizes the runtime of each stage.

The results indicates that the computational cost is dominated by neural reconstruction and point-cloud extraction, rather than by downstream geometric reasoning stages. NeRF optimization and point-cloud export account for most of the post-acquisition processing time, whereas RANSAC berry detection, stem skeletonization, and berry orientation optimization together require less than one minute. Including image acquisition, the system requires approximately 3 min 53 s before robot execution begins, followed by approximately 8.5 to 10 s of motion planning and approach execution per selected berry. This runtime is acceptable for experimental validation and for workflows in which the robot first scans a target bunch and then performs thinning decisions after reconstruction. However, real production deployment would require reducing idle time during reconstruction. A practical strategy is to pipeline sensing and computation across nearby bunches, while reconstruction and pose estimation are performed for one scanned

Table 9
Quantitative comparison of orientation optimization errors.

Representation	Mean (°)	Max (°)	Std (°)
AxisA	41.03	110.80	25.57
Quaternion	0.10	0.11	0.01
Euler	0.34	10.59	1.60
5D	1.94	34.02	6.19
6D	**0.09**	**0.11**	**0.01**

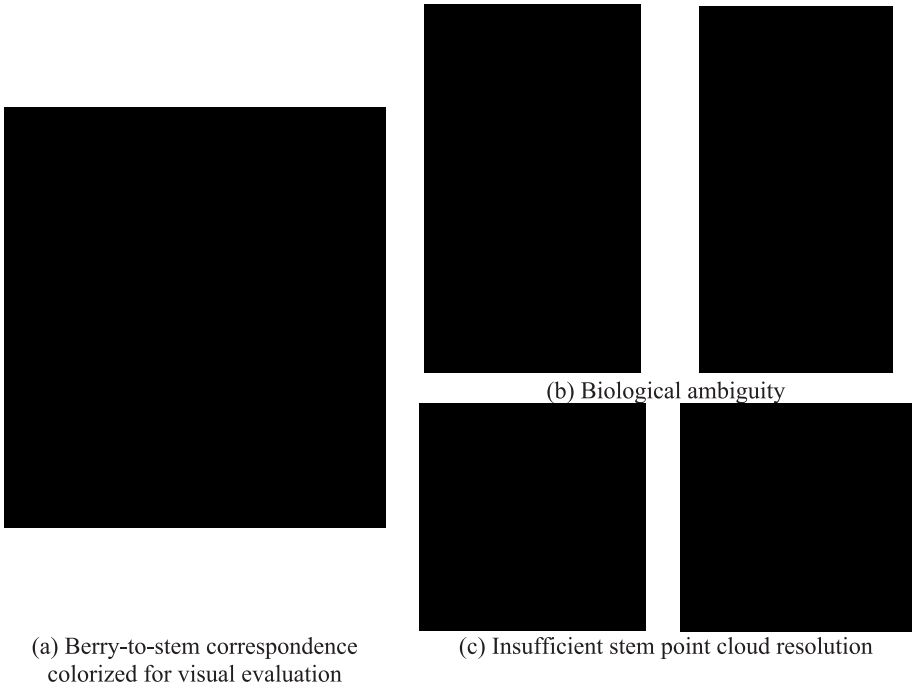


Fig. 23. Berry-to-stem topological assignment and failure modes. (a) Berry point clouds colorized by assigned stem for visual quality inspection. (b) Biological ambiguity: a berry physically overlaps a neighboring cluster (red circle), causing incorrect branch assignment. (c) Resolution failure: a missing fine sub-branch causes the berry (orange dot) to be assigned to an incorrect stem segment (orange arrow) instead of its true parent (blue arrow).

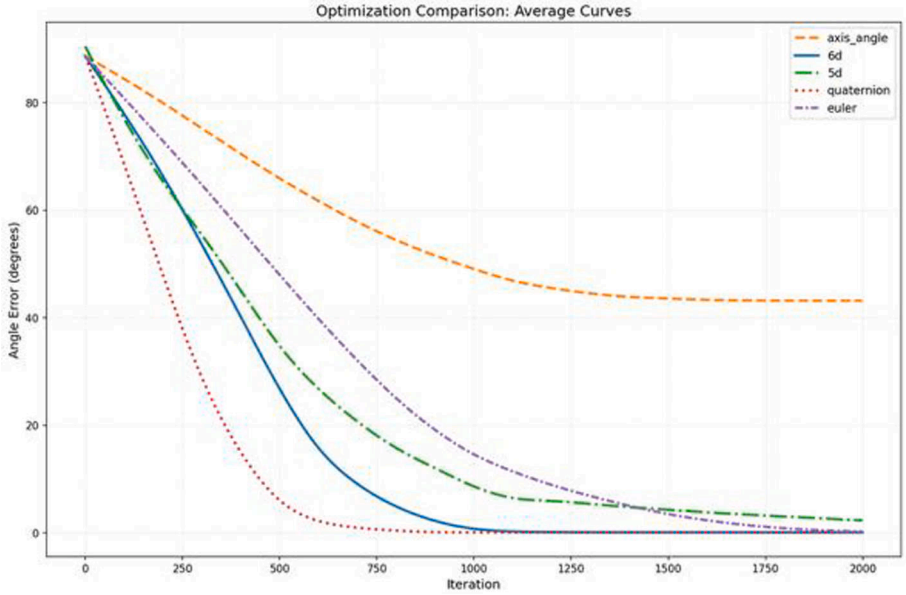


Fig. 24. Convergence of average orientation error during orientation optimization.

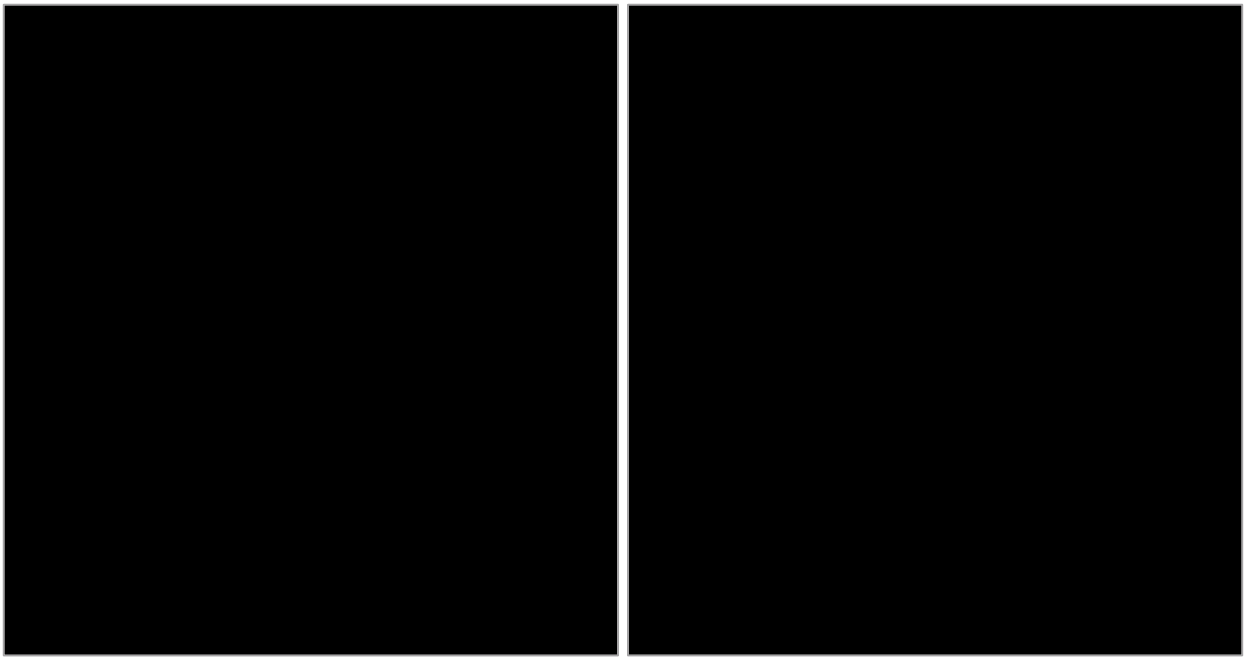


Fig. 25. Qualitative comparison of optimized end-effector orientations using (a) axis angle and (b) 6D rotation representations. For each detected berry, a local coordinate frame is visualized at the estimated target pose (red, green, and blue arrow denote the orientation of the end-effector frame, respectively), overlaid on the reconstructed grape bunch. Axis angle frequently yields inconsistent orientations across neighboring target, whereas the 6D representation produces more coherent and stable rotations relative to the local stem points, consistent with its lower optimization residuals.

bunch, image acquisition for the next bunch can be performed in parallel.

4. Limitations and future work

The proposed system demonstrates an end-to-end perception-to-manipulation pipeline for robotic grape berry thinning, but several limitations remain before production deployment. First, the current implementation requires a posed multi-view scan of one target bunch at a time. The view-sampling ablation indicates that overly sparse angular coverage can degrade reconstruction quality, berry counting, and berry localization, which may affect downstream thinning decisions such as local spacing estimation and berry removal selection. Therefore, the present system prioritizes dense 360° acquisition for reliable berry-level reconstruction. Although this target-bunch-level formulation is appropriate for robotic thinning, production environments often contain multiple neighboring bunches, leaves, and canopy structures. In this study, these surrounding structures were intentionally removed using spherical region-of-interest filtering, and multi-bunch or canopy-level reconstruction was not fully evaluated. Future extension could



Fig. 26. Metric consistency confirmation of exported point cloud. (a) Reference object with a known physical height 45 mm. (b) The reference object in 3D space dimension showing the measured height of 45.9 mm, demonstrating that the reconstruction preserves metric scale inherited from robot-provided camera poses and is directly usable for motion planning.

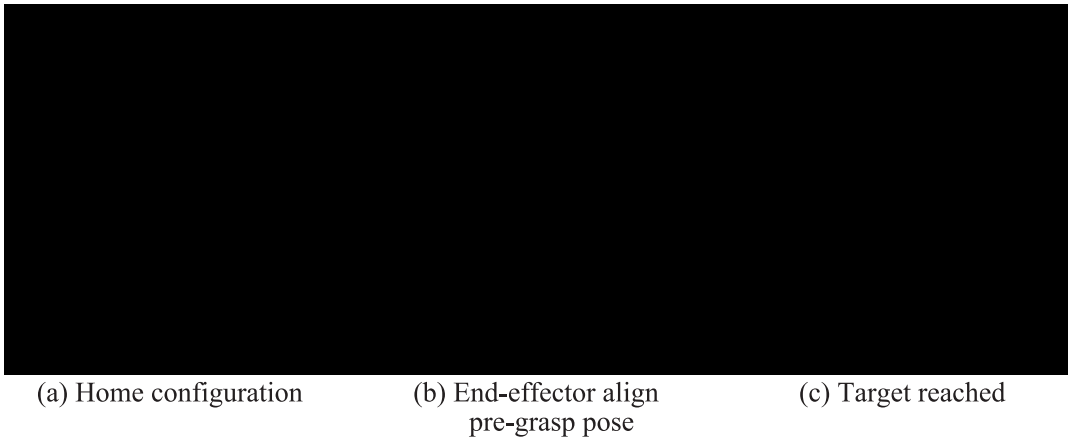


Fig. 27. Robot manipulation task decomposition for manipulation experiments. Sequential stages of 6-DoF berry approaching using Fanuc CRX-5iA collaborative robot. (a) Home configuration: robot positioned near grape bunch with end-effector in neutral orientation. (b) Pre-grasp pose: robot moved to 80 mm offset from target berry center along optimized approach vector (blue arrow). (c) Target pose reached: robot advanced final 80 mm along approach vector to berry surface contact position.

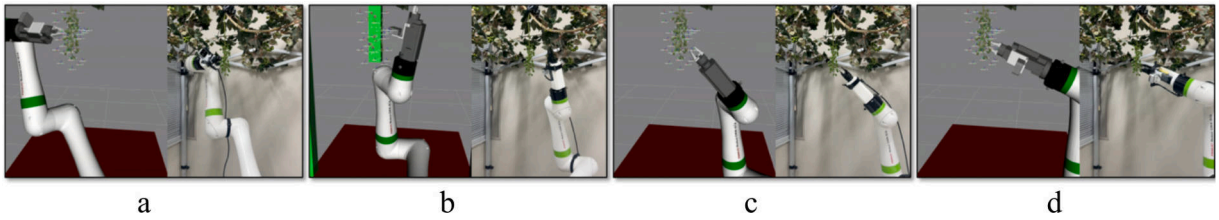


Fig. 28. Sequential trajectory visualization for representative berry approaching task. Four key waypoints showing robot motion from current configuration (a) to final target berry pose (d). The robot navigates collision-free through the cluttered grape canopy (visible as green point cloud) and support structures while maintaining end-effector orientation constraint. Sequential frames illustrate: (a) current configuration, (b-c) intermediate waypoints navigating around obstacles, and (d) final approach with aligned gripper orientation at target berry.

combine target-bunch reconstruction with bunch-level and leaf-level segmentation to support multi-bunch or canopy-level scene understanding. Such scene-level segmentation would be useful for collision detection, scan-path planning, and selecting the next target bunch during sequential robotic thinning.

Runtime is another remaining limitation, as per-bunch NeRF optimization and point-cloud export dominate the scan-to-pose processing time. Future work will investigate accelerated reconstruction, adaptive view planning, more efficient point-cloud extraction, and scan-process-return operation across nearby bunches. In such a workflow, the robot



Fig. 29. Robotic manipulation performance evaluation on 35 sequential berry targets. (a) Stage-wise reachability following task decomposition in Fig. 27. Left bar (blue): indicates successful motion from the initial state to the pre-grasp pose, Middle bar (orange): indicates pre-grasp poses reachability from previous configuration. Right bar (green/red): Green color indicates success case where IK solution is solved, and gripper aligned with target berry while red color indicates failure cases where no IK solution to approach target berry. (b) Execution time distributions for successful trials.

Table 10
Ablation study comparing approach orientation strategies on 35 sequential targets. Values report absolute counts and percentages. Collision-free insertion is evaluated only for trials that reached the target pose.

	Pre-grasp reachable	Target reachable	Success grip
Without orientation	35/35 (100%)	23/35 (65.71%)	18/23 (78.26%)
With orientation	30/35 (85.7%)	25/30 (83.33%)	24/25 (96%)

Table 11
Runtime breakdown of the proposed pipeline for one target grape bunch, including multi-view acquisition, NeRF reconstruction, point-cloud processing, berry-level pose estimation, and robot approach execution.

Processing stage	Time
Data acquisition	26 s
NeRF reconstruction	2 min 47 s
Point cloud export	35 s
RANSAC berry detection	5 s
Stem skeletonization	12 s
Berry orientation optimization	5 s
Robot approach per berry	8.5–10 s

can scan one bunch, move to acquire data from a neighboring bunch while reconstruction and pose estimation are performed for the previous bunch, and return to execute the planned thinning motion once processing is complete. This strategy could reduce idle time between image acquisition and robot execution while preserving the geometric completeness required for reliable thinning decisions.

In addition, some manipulation failures were caused by IK infeasibility under strict 6-DoF stem-aligned target poses rather than perception errors. This limitation could be mitigated by improved robot base placement, operation-zone selection, workspace-aware scan planning, or redundant manipulators such as 7-DoF arms. Finally, berry-to-stem assignment remains sensitive to missing fine pedicels under occlusion or insufficient reconstruction resolution, and broader validation is needed across cultivars, growth stages, illumination conditions, canopy densities, and full thinning operations, including berry selection, physical removal, and post-thinning quality assessment.

5. Conclusions

This paper presents an integrated perception-to-manipulation pipeline for robotic grape berry thinning, combining NeRF-based 3D reconstruction, RANSAC geometric segmentation, and structure-aware pose optimization. Three principal findings emerge from the evaluation.

First, deriving camera poses directly from robot forward kinematics via ROS 2 TF2 eliminates the scale ambiguity inherent in COLMAP-based NeRF workflows, producing metric-coordinate point clouds in the robot base frame without additional calibration. Scale verification confirmed sub-millimeter accuracy (45.9 mm reconstructed vs. 45.0 mm reference), establishing that reconstructed berry positions are directly usable for motion planning.

Second, RANSAC sphere fitting on NeRF-exported point clouds achieves 92.1% average precision and MAE of 1.1 berries, outperforming the learning-based SoftGroup++ baseline (82.8% AP) and reducing counting error by 52% compared to state-of-the-art 2D methods, without requiring any annotated training data. The geometry-based approach generalizes across bunch densities and berry sizes where learned methods exhibit systematic under-segmentation.

Third, stem-aligned 6-DoF pose optimization yields an end-to-end grip success rate of 68.6%, compared to 51.4% under a fixed-direction baseline — an absolute improvement of 17.2 percentage points. The ablation study reveals that orientation constraints reduce pre-grasp reachability by 14.3% due to stricter IK requirements, but this cost is recovered through a substantially higher collision-free grip rate (96% vs. 78.3%), confirming that structure-aware orientation estimation is essential for reliable manipulation in dense grape clusters.

The primary remaining limitation is IK infeasibility under strict 6-DoF orientation constraints, which accounts for all target-stage failures. This stems from optimizing only the approach axis direction while leaving gripper roll unconstrained, occasionally producing wrist configurations near joint limits. Joint-limit-aware pose optimization that jointly considers approach direction and wrist configuration represents the most direct path to improving end-to-end success. Computationally, NeRF training requires approximately 167 s per bunch; acceleration via Instant-NGP or 3D Gaussian Splatting is promising but requires careful validation to ensure sub-millimeter geometric accuracy is preserved, as manipulation success is sensitive to precise berry localization.

Beyond these technical improvements, broader deployment requires

validation across grape varieties, growth stages, and multi-season field conditions. The persistent 3D models produced by NeRF reconstruction also offer a natural path toward digital twin applications, where repeated reconstruction of the same bunch across growth stages could support predictive models of berry development and enable thinning decisions informed by forecasted cluster morphology rather than current state alone.

CRediT authorship contribution statement

Kabin Yok-In: Writing – original draft, Visualization, Validation, Software, Methodology. **Prawit Buayai:** Validation, Data curation, Conceptualization. **Dear Moeurn:** Validation, Software. **Xiaoyang Mao:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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